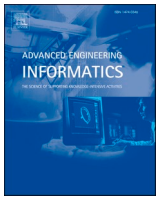
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A hybrid model for tool wear monitoring via physics-based and data-driven utilizing unscented Kalman Filter

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ABSTRACT

Tool Wear Monitoring (TWM) is crucial for ensuring part machining quality in metal cutting processes, with current prediction methods broadly categorized into three approaches: physics-based models, data-driven models, and hybrid models. While physics-based prediction methods are prone to significant modelling errors and low accuracy, data-driven approaches, despite achieving high accuracy, heavily depend on the quality of input data and exhibit insufficient reliability under complex operating conditions. Existing hybrid models predominantly emphasize single or specific tool wear mechanisms, which limits their adaptability to diverse and complex practical demands, hindering comprehensive and robust tool wear prediction. A hybrid tool wear prediction model integrating physics-based and data-driven models is proposed in this paper to address the aforementioned challenges. Firstly, the physics-based model is constructed based on primary wear mechanisms of abrasion, adhesion, and diffusion. Then, the TWM signals undergo preprocessing and multi-domain feature extraction. Following that, Spearman Correlation Coefficient (SCC)-based feature selection is applied to eliminate interference and redundancy, thereby enhancing the efficiency and accuracy of monitoring. Furthermore, data-driven model utilizing BiLSTM-Attention Mechanism (BiLSTM-AM) is constructed. Subsequently, the Unscented Kalman Filter (UKF) is employed to progressively and effectively integrate the predictions from both the physics-based model and the data-driven model, thereby achieving more precise tool wear predictions. Finally, we performed milling experiments across diverse operating conditions to demonstrate the performance and accuracy of the proposed method. The analysis of the experimental data reveals that, compared with using the physics-based model or the data-driven model independently, the proposed hybrid model achieves a reduction in the average RMSE by 46.5% 80.45%, and 55.11%, as well as a decrease in the average MAE by 46.75%, 80.90% and 57.22%. Furthermore, it enhances the R^2 values to 99.15%, 99.31% and 98.85% for the three datasets, respectively.

1. Introduction

As the end effector of machine tools, the cutting tool is the most susceptible to wear and failure within the machining system. The extent of its wear directly impacts critical aspects of the machining process, such as product quality, operational efficiency, energy consumption, and overall cost. During the cutting process, the continuous friction between the tool and the workpiece, as well as between the tool and the chips, inevitably leads to tool wear or damage [1]. This phenomenon is particularly pronounced when machining difficult-to-cut materials, as the wear rate of the tool is significantly accelerated. As the tool undergoes substantial wear, the rise in cutting forces, vibration, and cutting temperature can negatively impact the dimensional accuracy and surface quality of the workpiece [2]. In extreme scenarios, severe tool failure may result in scrapped parts, machine damage, and other significant losses [3]. Adopting a conservative tool replacement strategy could potentially lead to the underutilization of the remaining useful life (RUL) of tools and unnecessary downtime caused by premature tool changes. Data indicates that the actual useful life of tools utilized is only between 50% and 80% of their expected lifespan [4], the downtime for tool changes accounts for more than 20% of the total machine downtime [5], which subsequently increases production cycles and overall costs. By continuously monitoring tool states and employing predictive analytics for tool wear, tools can be replaced in a timely manner, thereby

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ensuring the quality, precision, and efficiency of the machining process. The implementation of Tool Wear Monitoring (TWM) systems in machine tools has been demonstrated to decrease production costs by 10% to 40% through the reduction of machine downtime and the optimization of the utilization of RUL [6]. Therefore, accurately predicting tool wear during the cutting process and enabling timely tool replacement before failure is of significant engineering importance for ensuring seamless production and improving product quality.

Over the recent years, numerous studies have focused on systematically enhancing the effectiveness and precision of TWM, resulting in the emergence of diverse tool wear prediction methods. These methods can be classified according to their modeling strategies into three primary categories: physics-based models [9–14], data-driven models [15–20] and hybrid models [21–24]. The prediction methods based on physics-based models and data-driven models continue to encounter limitations in terms of prediction accuracy and applicability. This is primarily owing to the intrinsic randomness of tool wear, the complexity of wear mechanisms, challenges associated with acquiring high-quality model training data, and the strong dependence of prediction accuracy on data quality. Prediction methods based on hybrid models have the potential to integrate the advantages of both physics-based and data-driven approaches. However, existing hybrid methods are still relatively limited and often necessitate the development of appropriate physics-based models that align with the characteristics of data-driven models and the specific research objectives. This complexity restricts their extensive application across diverse domains. Moreover, current research on physics-based data fusion for tool wear prediction predominantly emphasizes the influence of a single or specific physical factor, thereby imposing certain limitations. Consequently, the effective integration of multi-factor physics-based models into data-driven models continues to be a significant area warranting further investigation.

To tackle the aforementioned challenges, this paper proposes a hybrid tool wear prediction model based on physics-based and data-driven utilizing the Unscented Kalman Filter (UKF). This approach is designed to overcome the existing research bottleneck by integrating the strengths of both the physics-based model and data-driven model, thereby enhancing the accuracy of tool wear predictions. In contrast to the model that relies on data, the proposed method demonstrates superior generalization capabilities, while achieving higher fitting accuracy than the physics-based model. The main contributions of this research can be outlined as follows:

(1) This study proposes an innovative hybrid model based on the UKF, which integrates the developed physics-based model with the constructed data-driven model for high-precision dynamic estimation of tool wear states. The key innovation lies in the fact that it does not simply perform a static weighted average of the results of physics-based model and data-driven model. By incorporating physics-based wear increments as state equation priors and data-driven model outputs as observations, the UKF establishes a dual-model collaboration mechanism. During the prediction phase, nonlinear dynamic characteristics are captured through unscented transform, while the Kalman gain dynamically adjusts the weighting between the two models in the update phase, thereby achieving accurate tool wear state estimation. Through comprehensive experimental verification, we have not only demonstrated that this hybrid model significantly outperforms any single model in terms of prediction accuracy, but more importantly, revealed its excellent generalization ability and robustness when facing unknown cutting conditions. This hybrid model significantly enhances system adaptability to dynamic machining conditions while preserving physical interpretability, providing a novel approach for addressing nonlinear state estimation challenges in manufacturing processes.
(2) The physics-based model for the flank wear of end mills, incorporating abrasive, adhesion, and diffusion wear mechanisms, are employed to provide a clear representation of the physical phenomena during cutting. It represents a more comprehensive physics-based model that integrates multiple wear mechanisms, as compared to models based on a single mechanism. Its strong generalization capability allows application across materials and operating conditions while enabling deeper insight into wear mechanisms. Therefore, the physics-based model better aligns with real-world industrial processing conditions and provides a more robust foundation for subsequent hybrid modeling.
(3) Spearman Correlation Coefficient (SCC) is utilized to perform feature selection on the multi-domain feature matrix, effectively diminishing the matrix's complexity while preserving the nonlinear correlations within the data. This methodology ensures minimal loss of critical information, thereby substantially enhancing the accuracy of tool wear prediction and providing robust support for the data-driven model.

The rest of this paper is structured in the following manner. Section 2 provides a comprehensive review of existing approaches for tool wear prediction. Section 3 outlines the proposed method, offering a detailed explanation of the overall framework and the specific steps involved. Section 4 introduces the experimental setup and data processing. In Section 5, a thorough analysis of the comparison results and model performance are presented. Section 6 presents the concluding remarks, summarizing the key findings and offering recommendations for future research avenues.

2. Review of related work

2.1. The prediction methods based on physics-based models

The prediction method based on physics-based models examines the tool wear degradation process by leveraging physical principles and mathematical methodologies to construct precise tool wear models. This approach provides robust interpretability while requiring a minimal amount of data. Physics-based models can be classified into two categories: analytical models [7] and empirical models [8]. These models predict tool wear from different perspectives, specifically focusing on wear mechanisms and degradation patterns, respectively. Table 1 provides a comprehensive summary of prediction methods based on physics-based models.

The analytical model leverages physical prior knowledge to systematically analyze the tool wear process, with a focus on quantifying the wear rate under the synergistic effects of multiple wear mechanisms to enable accurate tool wear prediction. Yang et al. [9] combined abrasive, diffusion, and adhesive wear mechanisms to predict flank wear in titanium alloy turning. Liu et al. [10] linked cutting force coefficients to tool wear estimation. Liu et al. [11] constructed a dynamic prediction model derived from micro-milling forces. Additionally, Pálmai et al. [12] proposed a wear rate model incorporating cutting distance and temperature for lifetime estimation.

The empirical model is developed based on practical cutting experience and extensive experimental data, establishing a systematic empirical relationship between tool parameters and tool wear. Zhu et al. [13] proposed an adjustable-coefficient model to enhance adaptability in high-speed milling. To address the issue that most tool wear models fail to incorporate wear rate and only focus on final wear (i.e., limit wear), such as Laakso et al. [14] introduced a logarithmic function to describe dynamic wear rates.

However, physics-based models also possess inherent limitations. The tool wear process is inherently dynamic and highly complex. In the analysis and physical modeling of this process, numerous approximations and simplifications are inevitably required. Establishing an accurate model that fully satisfies practical requirements is challenging, which in turn results in reduced prediction accuracy for tool wear.

Table 1
A summary of prediction methods based on physics-based models.

Ref.	Method/Model	Core Technology/ Algorithms	Key Contributions
Yang et al. [9]	Multi-mechanism wear model	Abrasive, diffusion, and adhesive wear mechanisms	Predicted flank wear by combining three wear mechanisms for titanium alloy machining
Liu et al. [10]	Cutting force model	Axial depth of cut and feed rate analysis	Linked cutting force coefficients to tool wear estimation
Liu et al. [11]	Micro-milling force model	Variable cutting parameters, tool runout, and wear effects	Established a wear prediction model based on milling force coefficients
Pálmai et al. [12]	Tool wear rate model	Cutting distance, diffusion wear, oxidation wear	Enhanced adaptability to milling conditions and predicted tool lifecycle wear
Zhu et al. [13]	Adjustable-coefficient empirical model	General tool wear relationships with tunable parameters	Optimized tool wear prediction performance through parallel deep learning structures combined with attention mechanisms
Laakso et al. [14]	Logarithmic wear rate model	Time-dependent wear rate variation	Predicted dynamic wear progression and final wear limits under specific conditions
In the hybrid model, we adopt the physics-based model.	Analytical model for the flank wear of end mill	− Multi-mechanism wear (abrasive, adhesive, diffusion) − Stress-temperature coupling (Eqs. 1–4) − Geometric-mechanistic volume equivalence (Eqs. (8)–(11)	1. Multi-mechanism modeling: Captures the interactions of abrasive, adhesive, and diffusive wear together. 2. Stress-temperature coupling: Combines stress and temperature to calculate wear rate. 3. Geometry-Mechanism Consistency: Ensures the wear volume from mechanisms matches the geometric wear volume, allowing the creation of a dynamic model for flank wear rate.

2.2. The prediction methods based on data-driven models

The data-driven prediction approach analyzes sensor data (e.g., force, vibration signals) that are collected during the cutting operation, thereby establishing a correlation between TWM signals and tool wear labels. The general process of the data-driven method is shown in Fig. 1. Compared with physics-based models, data-driven approaches are typically more straightforward in terms of comprehension and implementation. With the remarkable progress made in the fields of machine learning (ML) and deep learning (DL), these methodologies have been extensively adopted across various domains and have consistently exhibited superior predictive performance.

ML has become the predominant methodology for developing tool wear prediction models, owing to its superior data processing capabilities. Early ML approaches predominantly depended on single sensor signals, which inherently possess certain limitations. Karabacak et al. [15] employed a multi-sensor fusion strategy that integrated data from vibration, acoustic emission, and motor current sensors. This data was analyzed using machine learning techniques like artificial neural networks (ANN), support vector machines (SVM), and Gaussian process regression (GPR). This methodology markedly improved the system's accuracy and reliability. Jia et al. [16] employed Extreme Randomized Trees (ET) for feature scoring and screening, utilizing the Pearson Correlation Coefficient (PCC) to evaluate feature-target vector correlations and select strongly relevant features. They subsequently created a predictive model for tool wear utilizing the Relevance Vector Machine (RVM). This dual-feature selection approach effectively preserves sensitive feature information while enhancing both prediction efficiency and accuracy.

With the decreasing costs of data acquisition and computation, DL has experienced rapid advancement in the manufacturing domain, particularly in tool wear prediction [17]. In this context, DL enables direct mapping between monitoring signals and tool wear states. For instance, Cheng et al. [18] proposed a hybrid framework that integrates feature normalization and attention mechanisms to mitigate the influence of cutting condition dependencies. This framework is further enhanced by parallel CNN-BiLSTM networks, which significantly improves the precision of the model. Hou et al. [19] addressed the challenge of cross-domain signal disparities by employing a CNN-LSTM model augmented with domain-adversarial adaptation and channel attention mechanisms, thereby improving the model's generalization capability. Additionally, Duan et al. [20] introduced a Parallel Deep Learning framework incorporating Hybrid Attention (HABPDL) mechanisms, which integrates CNN-RNN blocks with hybrid attention mechanisms to enhance predictive performance. A summary of prediction methods based on data-driven models is provided in Table 2.

However, data-driven models also face several significant challenges. Their prediction accuracy is heavily contingent upon both the data's quality and its quantity, which typically necessitates the collection of large datasets through expensive and time-intensive measurements. During the process of sensor data acquisition, external factors such as environmental conditions may substantially influence the accuracy and dependability of the collected data. As a result, insufficient data or high levels of noise can compromise the generalization capability of data-driven models. Furthermore, these models often function as black-box systems, leading to a lack of interpretability and complicating the understanding of their underlying decision-making mechanisms.

2.3. The prediction methods based on hybrid models

To overcome the shortcomings of traditional physics-based models, data-driven models, such as insufficient accuracy, inadequate training data, or limited interpretability, recent artificial intelligence research has focused on integrating data-driven methodologies with physical knowledge. This approach enhances model interpretability and robustness while overcoming associated challenges. Notably, this integration significantly improves the reliability and precision of tool wear prediction models.

For instance, Zhu et al. [21] introduced a physics-based Hidden Markov Model (PI-HMM), which integrates empirical coefficients with physically meaningful labels. This approach effectively addresses the issue of data scarcity and improves the generalization capability of the model. Li et al. [22] integrated Gaussian Process Regression (GPR) with wear mechanisms by incorporating feature sequence numbers and linear constraints, thereby enhancing prediction accuracy under complex conditions. Wang et al. [23] integrated the empirical formula into a neural network framework, leveraging a physics-based loss function to address prediction inconsistencies. Hanachi et al. [24] fused the Adaptive Neuro-Fuzzy Inference System (ANFIS) with the empirical wear-time model proposed by particle filtering, thereby facilitating probabilistic time-series predictions and effectively reducing prediction errors. A summary of prediction methods based on hybrid models is shown in Table 3.

Through the above analysis, we can find that the advantages of hybrid models can be summarized from two key perspectives: 1) By incorporating mechanistic information, these models are confined to a low-dimensional space. Even when the available samples are insufficient to encompass all operational conditions, the model is still capable of

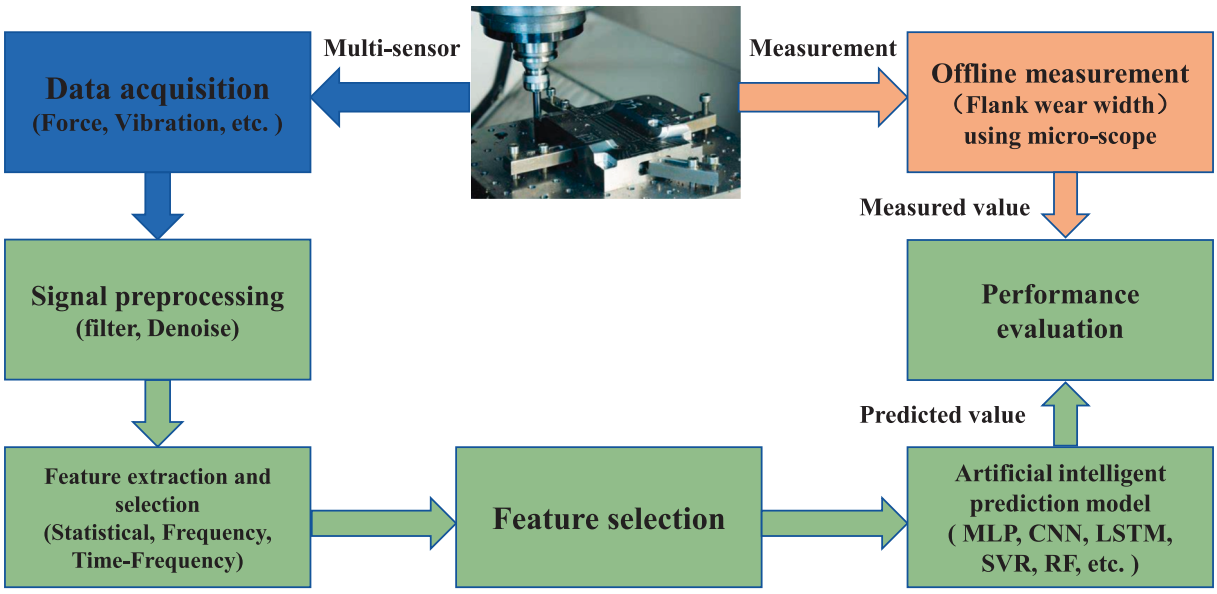


Fig. 1. The general process of using data-driven methods to monitor tool wear.

Table 2
A summary of prediction methods based on data-driven models.

Ref.	Method/Model	Core Technology/ Algorithms	Key Contributions
Karabacak et al. [15]	Multi-sensor signal fusion	ANN, SVM, GPR	Improved prediction accuracy and robustness through multi-sensor fusion
Jia et al. [16]	ET-PCC-RVM method	Extreme Random Trees (ET), Pearson Correlation Coefficient (PCC), RVM	Combined feature selection and correlation analysis to avoid information loss and improve prediction efficiency/accuracy
Cheng et al. [18]	Feature normalization + Attention mechanism + Deep learning framework	Parallel CNN, BiLSTM, Attention mechanism	Eliminated dependency of features on cutting conditions, enhanced key information capture, and improved TWM accuracy
Hou et al. [19]	CNN-LSTM network	Domain-adversarial adaptation, Channel attention mechanism	Addressed cross-domain signal distribution differences and enhanced model generalization ability
Duan et al. [20]	HABPDL hybrid model	CNN-RNN hybrid structure, Hybrid attention mechanism	Optimized tool wear prediction performance through parallel deep learning structures combined with attention mechanisms
In the hybrid model, we adopt the data-based model.	Bidirectional LSTM with Multi-Head Attention	− BiLSTM − Multi-head attention mechanism − Sensitive feature selection based on SCC.	1. Captures bidirectional temporal dependencies (past/future context) via BiLSTM. 2. Adaptive multi-perspective attention dynamically focuses on critical wear stages.3. SCC-based feature selection eliminates noise and enhances input relevance.

generalizing effectively to unseen scenarios. 2) The integration of domain knowledge ensures that hybrid models not only exhibit interpretability but also conform to physical principles, thereby aligning with the modern manufacturing industry's focus on reliability and well-established technologies [25].

However, existing hybrid approaches still encounter the following challenges: 1) The incorporated mechanism information tends to be oversimplified, and the full potential of complex mechanism information has yet to be effectively harnessed; 2) Current hybrid approaches remain relatively constrained, as most methods are designed for specific mechanisms, thereby posing significant challenges to cross-domain applications; 3) A substantial portion of the existing research predominantly focuses on the integration of single mechanisms with data-driven models, while the effective fusion of multiple physical factors remains a largely unresolved challenge.

In conclusion, relying exclusively on prediction methods based either on physics-based models or data-driven models for tool wear prediction often presents significant limitations. The hybrid prediction approach can efficiently harness the strengths of both approaches by incorporating information from diverse sources, thereby enhancing prediction accuracy and applicability. However, most current research focuses on the fusion of single or specific mechanisms, which makes it difficult to meet practical application needs. To overcome these challenges, this study proposes a method based on UKF, which effectively integrates the data-driven model with the physics-based model for the flank wear of end mills caused by abrasive, adhesive, and diffusive mechanisms. This method dynamically updates the tool wear state, improving the accuracy and reliability of tool wear prediction.

3. Research methodology

This paper presents an overall framework for tool wear prediction by integrating a physics-based model and a data-driven model based on UKF, as illustrated in Fig. 2. The proposed method comprises four primary components: construction of the physics-based model, data processing, development of the data-driven model, and hybrid model. Firstly, the physics-based model is established by an analytical model for end mill flank wear prediction, incorporating abrasive, adhesive, and diffusive wear mechanisms. Secondly, the collected data is preprocessed and features are extracted to provide foundational support for data-driven model development. Thirdly, the data-driven model is conducted by BiLSTM-AM after feature selection from raw features via SCC. Finally, the hybrid tool wear prediction model is implemented through

Table 3
A summary of prediction methods based on hybrid models.

Ref.	Method/Model	Core Technology/Algorithms	Key Contributions
Zhu et al. [21]	Physical Information Hidden Markov Model (PI-HMM)	Adjustable empirical model and physical information fusion	Overcame label scarcity, improved interpretability and generalization by integrating physics and data-driven methods
Li et al. [22]	Gaussian Process Regression with Wear Mechanisms	Feature sequence numbers, GPR constraints	Enhanced prediction accuracy under varying conditions by combining GPR and wear mechanisms
Wang et al. [23]	Physically Guided Neural Network	Physical loss function, empirical formula:$\Delta\theta = CN^m$, where $\Delta\theta$ represents the increment of tool wear, C is the proportionality constant, N is the cutting force, and m is the index term of cutting force.	Eliminated physical inconsistencies in predictions via physics-guided training
Hanachi et al. [24]	ANFIS-SIPOS-Particle Filter Framework	ANFIS, the empirical wear-time model proposed by Siops, particle filtering	Reduced prediction error by serializing wear progression and representing results probabilistically
Our approach	Data and mechanism fusion using UKF	– BiLSTM-AM – Multi – mechanism physics-based model (abrasion, adhesion, diffusion)- UKF	1. Dynamic fusion: UKF adaptively weights predictions from the physial model and the data-driven model under varying conditions. 2. BiLSTM-AM modeling: Combines bidirectional temporal dependency learning (BiLSTM) with multi-head attention to dynamically weight critical wear-related features. 3. Multi-mechanism physical integration: Captures complex wear interactions (abrasion, adhesion, diffusion) beyond single-mechanism limitations.

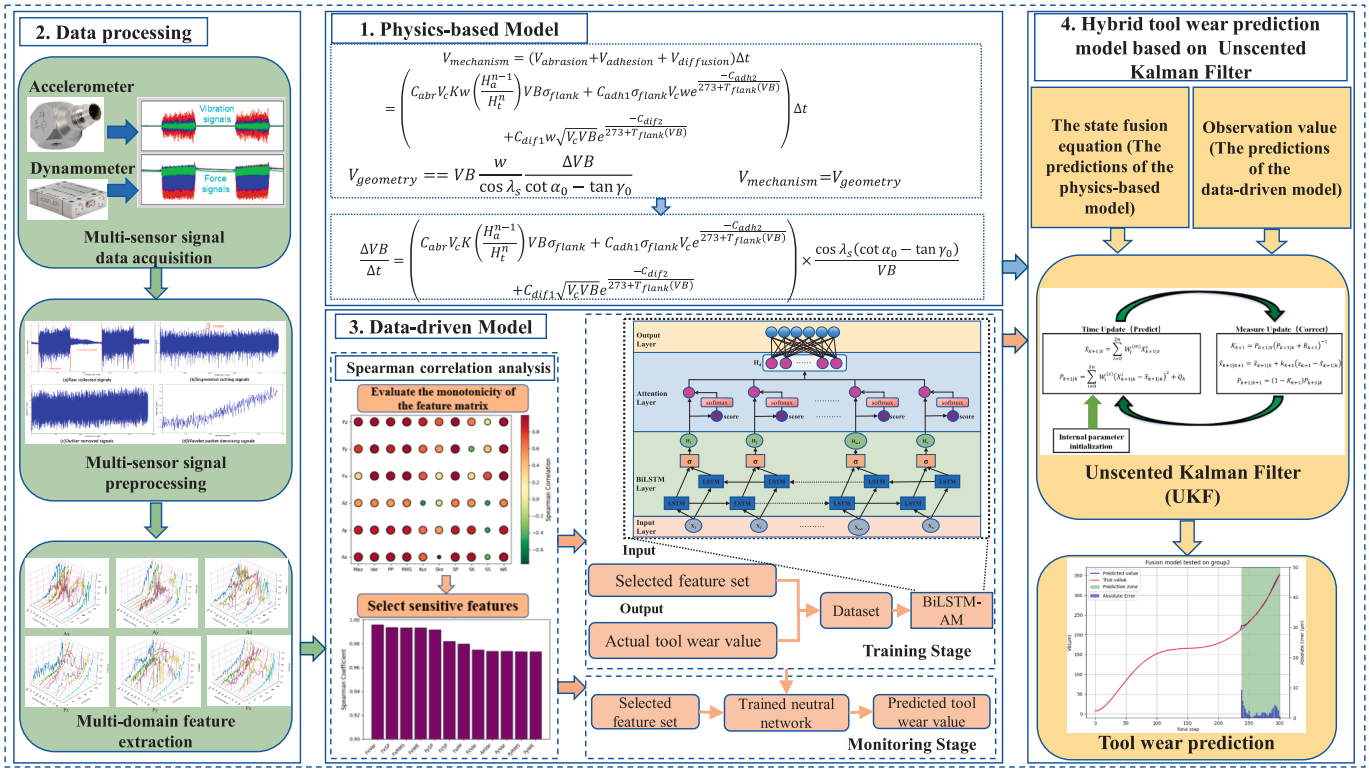


Fig. 2. The overall framework of the proposed method.

UKF approach to dynamically adjust the weight for prediction results both the physics-based model and the data-driven model.

3.1. Physics-based model

Compared to traditional empirical formulas, the physics-based model with analytical approach can more comprehensively consider the complex physical phenomena during the cutting process. Accordingly, the analytical method is employed in this research to examine the flank wear of end mills [26]. To enhance the accuracy and applicability of tool wear prediction, this model incorporates three primary wear mechanisms: abrasive, adhesive, and diffusion wear. It not only has stronger generalization ability, making it applicable to different materials and operating conditions, but also provides a deep understanding of the tool wear mechanisms. It offers theoretical support for optimizing cutting parameters and improving tool remaining useful life. The modeling process is shown in Fig. 3.

This model employs a multi-mechanism cross-scale framework. Using inputs of cutting parameters, tool geometry parameters, and material properties of tool and workpiece, it quantifies mechanical-thermodynamic states via flank normal stress and temperature field models. A dynamic model of flank wear of end mill integrates abrasion, adhesion, and diffusion mechanisms. Coupling mechanistic and geometric models enables dynamic prediction of the length of the flank

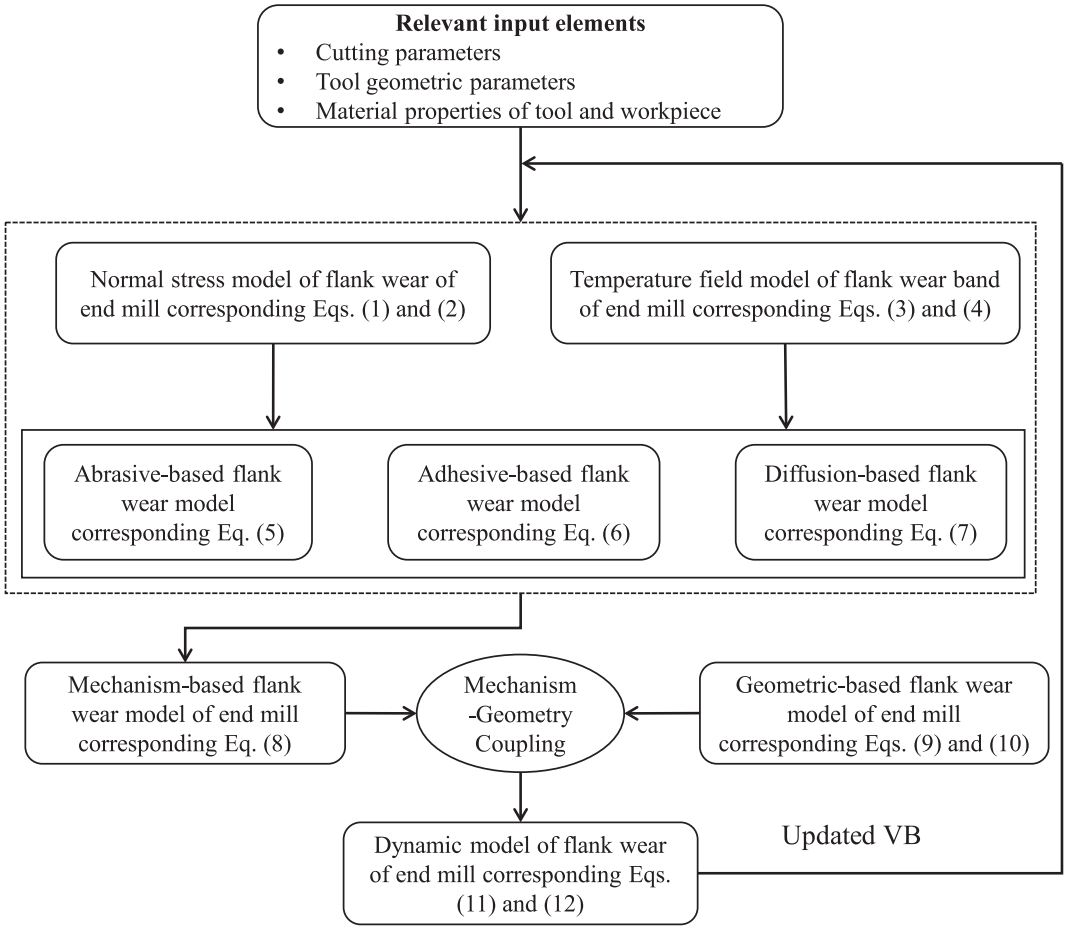


Fig. 3. The establishment process of the physics-based model [26].

wear band (VB) through iterative updates. This closed-loop from physical field to mechanism and geometry approach reveals stress/temperature-wear morphology interactions, offering an interpretable and applicable tool for end mill wear prediction.

3.1.1. Normal stress model and temperature field model of flank wear

In tool wear modeling, normal stress and temperature field are critical drivers: normal stress governs mechanical energy transfer in abrasive wear and adhesive wear, while temperature field accelerates diffusion and thermomechanical wear via material softening. Modeling these fields captures spatiotemporal stress and temperature evolution, enabling quantitative multi-mechanism wear analysis. This forms the theoretical basis for coupled wear modeling.

(1) Normal Stress Model

Based on the research conducted by Hou et al. [27], the normal stress σ_{flank} can be represented as:

$$\sigma_{flank}(x) = \begin{cases} \sigma_{flank-tip} & x \in (VB_p, VB) \\ \sigma_{flank-tip}\left(\dfrac{x}{VB_e}\right)^2 & x \in (0, VB_e) \end{cases} \# \tag{1}$$

Where $\sigma_{flank-tip}$ is the maximum normal stress near the tool tip on the flank; VB_p and VB_e represent the wear widths of the plastic and elastic contact regions on the flank, respectively.

$\sigma_{flank-tip}$ can be expressed as [28]:

$$\sigma_{flank-tip} = Y\left[1 + \frac{\pi}{2} - 2\rho - 2\phi + cos^{-1}(m_w) + sin\left(cos^{-1}(m_w)\right)\right] \# \tag{2}$$

Where Y represents the ratio of shear stress at the tool tip to that on the cutting surface, equating to the friction coefficient μ_c near the flank face cutting edge. In this study, the workpiece material is mold steel, so $\mu_c = 0.9$ [29]; ρ is the rake angle of the workpiece in front of the end mill, with a value of 0.8° [30]; m_w denotes the flank wear band's slip-line field angle, which is equal to μ_c; and ϕ is the shear angle.

(2) Temperature Field Model

Based on the research conducted by Moufki et al. [31], the temperature $T_{flank}(VB)$ can be represented as:

$$T_{flank}(VB) = \frac{7}{15}\frac{\mu_c \sigma_{flank-tip}\sqrt{V_c \cdot VB}}{\sqrt{\pi \rho_w c k}} + T_0 \# \tag{3}$$

Where k is the thermal conductivity of the workpiece; c is the specific heat capacity of the workpiece; ρ_w is the density of the workpiece; and T_0 is the initial temperature at the start time.

The relative slip velocity V_c can be expressed as:

$$V_c = V\frac{cos\lambda_s sin\phi}{cos\eta_c cos(\phi - \alpha_0)} \# \tag{4}$$

Where V is the cutting speed; λ_s is the cutting edge inclination; η_c is the chip flow angle; and α_0 is the clearance angle.

3.1.2. Abrasive-based flank wear model

Abrasive wear is caused by the repeated extrusion and friction between the cutting tool and the workpiece during milling. High temperatures lead to material softening, while hard particles (such as carbides) induce surface scratching, progressively wearing down the cutting tool.

As a result of the continuous and periodic intense contact during machining, abrasive wear accumulates gradually over time. When the workpiece material contains hard particles with a hardness comparable to that of the cutting tool, abrasive wear becomes the predominant wear mechanism. In homogeneous materials, however, the absence of hard particle cutting action results in an order-of-magnitude reduction in the contribution of this wear mechanism. Building on the established normal stress field model (Eqs. (1)–(2) and hardness ratio analysis, this section adapts an abrasive-based flank wear model (Eq. (5) through scratching energy–stress coupling quantification, establishing the mechanical wear component for multi-mechanism modeling.

Huang et al. [32] provided the calculation formula for the abrasive wear rate $V_{abrasion}$ during the milling process.

$$V_{abrasion} = C_{abr} V_c K w \left(\frac{H_a^{n-1}}{H_t^n}\right) VB\sigma_{flank} \# \tag{5}$$

Where C_{abr} is the abrasive wear coefficient, with a value of 2.37×10^{-11} [33]; w is the milling width; K and n are functions of H_a/H_t; H_t is the tool hardness, and since the tool material is high-speed steel, H_t = 66.5HRC; H_a is the abrasive hardness, given by the thermal hardness data [26], $H_a = \frac{H_0}{e^{aT_{flank}(VB)}}$, where $H_0 = 2350kg/mm$ and $a = 3.62 \times 10^{-4}$ °C.

3.1.3. Adhesive-based flank wear model

Adhesive wear is caused by the repeated extrusion and friction between the tool and the workpiece. At high temperatures (exceeding 600°C), the material in the contact region softens, leading to the formation of microscopic adhesion points. During the friction process, material is torn away and transferred, causing the built-up edge to detach. At high temperatures, materials exhibit increased susceptibility to adhesion and tearing, leading to a more pronounced alternating "adhesion – detachment" behavior during wear. This phenomenon is markedly amplified in highly ductile materials due to the continuous regeneration of the bonding layer. The extent of wear is exponentially correlated with the material's plastic deformation capability. Conversely, when processing hard materials, the predominance of the brittle peeling mechanism significantly suppresses the potential for bonding effect evolution. By integrating the temperature field distribution (Eq. (3) with stress gradients (Eq. (1), an adhesive-based flank wear model (Eq. (6) quantifies the interplay between thermal softening and adhesion failure. This formulation provides a physics-based component for multi-physics wear modeling.

The adhesive wear rate $V_{adhesion}$ is calculated using the formula proposed by Usui et al. [34]:

$$V_{adhesion} = C_{adh1}\sigma_{flank} V_c w e^{\frac{-C_{adh2}}{273+T_{flank}(VB)}} \# \tag{6}$$

Where C_{adh1} and C_{adh2} are the adhesive wear coefficients, with values of 7.80×10^{-9} and 2.50×10^{3} [35], respectively.

3.1.4. Diffusion-based flank wear model

Diffusion wear is caused by high temperatures (exceeding 800°C) during the machining process. Atoms in the tool material migrate to the workpiece driven by both high temperatures and the chemical potential difference between the materials. As cutting progresses, the loss of atoms leads to the formation of diffusion channels on the surface layer of the tool, causing the wear rate to increase sharply with rising temperature. In the processing of superalloys, this mechanism becomes the predominant wear mechanism due to the intense chemical interactions between the material properties. When machining conventional steel components, the reduced thermodynamic driving force and enhanced interface stability cause diffusion behavior to act as a secondary factor. Diffusion wear is modeled through temperature field analysis (Eq. (3). By incorporating activation energy theory, a diffusion-based flank wear model (Eq. (7) quantifies atomic diffusion rates as an exponential function of temperature. This formulation integrates thermodynamic driving forces into the multi-mechanism framework, improving thermal wear prediction accuracy.

The diffusive wear rate $V_{diffusion}$ is calculated using the formula proposed by Huang et al. [32]:

$$V_{diffusion} = C_{dif1} w \sqrt{V_c VB} e^{\frac{-C_{dif2}}{273+T_{flank}(VB)}} \# \tag{7}$$

Where C_{adh1} and C_{adh2} are the diffusion wear coefficients, with values of 2.64×10^{-3} and 6.86×10^{3} [36], respectively.

3.1.5. Dynamic model of flank wear of end mill

The dynamic model of flank wear of end mill integrates multi-physics mechanisms with geometric deformation through a "physical-geometric" coupling framework. Firstly, it quantifies the synergistic effects of abrasive, adhesive, and diffusive wear mechanisms based on spatio-temporal distributions of normal stress fields and temperature fields. Subsequently, cross-scale mapping relationships are constructed via equivalent conversion between the mechanism-driven volumetric wear model and the geometry-driven volumetric wear model of flank wear. Ultimately, the dynamic model of flank wear of end mill is derived, enabling time-variant iteration of VB values. This approach correlates stress/temperature gradients with wear morphology evolution while ensuring robustness in complex conditions through physical-geometric closed-loop feedback.

Assume that the flank wear of the end mill is solely attributed to the aforementioned three types of wear, and the three established wear models are mutually independent. The mechanism-driven volumetric wear model $V_{mechanism}$, represents the superposition of the three established wear volume models. Consequently, $V_{mechanism}$ can be expressed as:

$$\begin{aligned} V_{mechanism} &= (V_{abrasion} + V_{adhesion} + V_{diffusion})\Delta t \\ &= \left(\begin{array}{c} C_{abr} V_c K w \left(\frac{H_a^{n-1}}{H_t^n}\right) VB\sigma_{flank} + C_{adh1}\sigma_{flank} V_c w e^{\frac{-C_{adh2}}{273+T_{flank}(VB)}} \\ + C_{dif1} w \sqrt{V_c VB}\, e^{\frac{-C_{dif2}}{273+T_{flank}(VB)}} \end{array} \right) \Delta t \# \end{aligned} \tag{8}$$

Where Δt is the milling time.

The geometry-driven volumetric wear model is shown in Fig. 4.

The geometry-driven volumetric wear model $V_{geometry}$ can be expressed as:

$$\begin{aligned} V_{geometry} &= S_{ABCDEF} \cdot w = (S_{AEF} + S_{ABDE} + S_{BCD}) \cdot \frac{w}{cos\lambda_s} \\ &= \left[|AE|^2 tan\gamma_0 + VB|AE| + \frac{1}{2} cot\alpha_0 |AE|^2 \right] \cdot \frac{w}{cos\lambda_s} \# \end{aligned} \tag{9}$$

Where γ_0 is the rake angle.

The Taylor expansion excludes the higher-order term $|AE|^2$, given its contribution magnitude is under 3% (10^{-4} mm^2) when VB $\leq$ 0.3 mm, and its inclusion would induce nonlinear instability in the governing equations of the wear evolution model. Eq. (9) can be transformed into Eq. (10) as follows:

$$\begin{aligned} V_{geometry} &= VB|AE| \frac{w}{cos\lambda_s} = VB \frac{w}{cos\lambda_s} \frac{|AE|(cot\alpha_0 - tan\gamma_0)}{cot\alpha_0 - tan\gamma_0} \\ &= VB \frac{w}{cos\lambda_s} \frac{\Delta VB}{cot\alpha_0 - tan\gamma_0} \end{aligned} \# \tag{10}$$

Where ΔVB is the increment of the tool wear width in flank within the time interval Δt.

$V_{mechanism}$ and $V_{geometry}$ are approximately equal to Eq. (11).

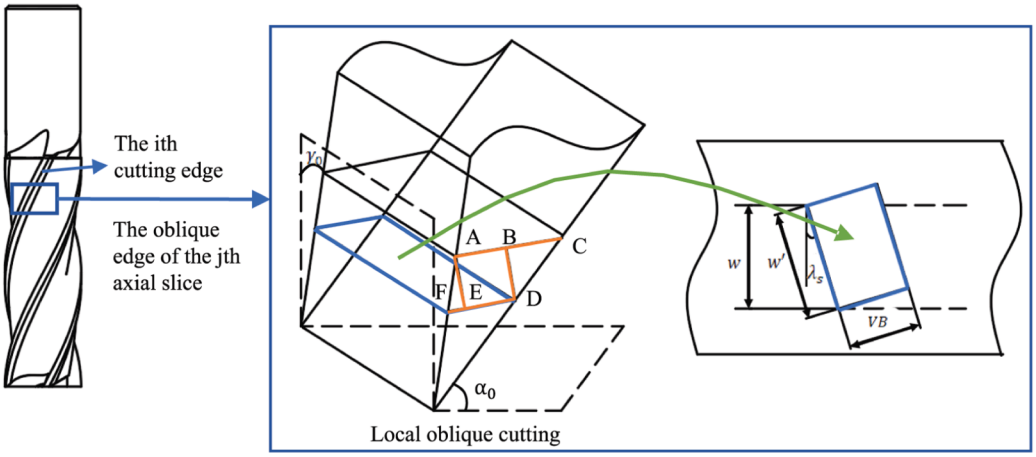


Fig. 4. Geometry-based volume model of flank wear [26].

$$VB\frac{w}{cos\lambda_s}\frac{\Delta VB}{cot\alpha_0 - tan\gamma_0} = \left(C_{abr}V_cKw\left(\frac{H_a^{n-1}}{H_t^n}\right)VB\sigma_{flank} + C_{adh1}\sigma_{flank}V_cwe^{\frac{-C_{adh2}}{273+T_{flank}(VB)}} + C_{dif1}w\sqrt{V_cVB}\,e^{\frac{-C_{dif2}}{273+T_{flank}(VB)}} \right)\Delta t \quad \# \quad (11)$$

Eq. (11) can be further derived to obtain the expression for the dynamic model of flank wear of end mill:

$$\frac{\Delta VB}{\Delta t} = \left(C_{abr}V_cK\left(\frac{H_a^{n-1}}{H_t^n}\right)VB\sigma_{flank} + C_{adh1}\sigma_{flank}V_ce^{\frac{-C_{adh2}}{273+T_{flank}(VB)}} + C_{dif1}\sqrt{V_cVB}\,e^{\frac{-C_{dif2}}{273+T_{flank}(VB)}} \right) \# \times\frac{cos\lambda_s(cot\alpha_0 - tan\gamma_0)}{VB} \quad (12)$$

Eq. (12) represents the final formulation of the multi-mechanism physics-based model developed in this study, which describes the rate of change of flank wear width *VB* with respect to time Δt. By incorporating the three primary wear mechanisms (abrasive wear, adhesive wear, and diffusion wear), the model provides a comprehensive characterization of dynamic process of tool wear evolution. The first three terms on the right-hand side correspond respectively to the contributions of different wear mechanisms. Each sub-term incorporates cutting parameters, material properties, and wear-related physical quantities, which provides a systematic quantification of each mechanism's influence on the total wear rate. The last term $\frac{\cos\lambda_s(\cot\alpha_0-\tan\gamma_0)}{VB}$ is the geometric correction coefficient. Based on the equivalence between mechanical wear volume and geometric wear volume, the volumetric wear rate is converted into a width-based wear rate to ensure consistency between the physical mechanism and the geometric representation. Furthermore, the model establishes a functional relationship between the wear rate and key variables such as *VB*, σ_{flank}, T_{flank}. It reveals the nonlinear dynamic characteristics of the wear process and serves as the state transition equation in the subsequent hybrid model to provide a physical basis for the real-time prediction of tool wear.

3.2. Data processing using multi-domain feature extraction and selection

3.2.1. Multi-domain feature extraction

During progressive tool cutting, multi-sensor data acquisition continuously expands, generating significant redundancy that necessitates additional processing. This redundant information complicates the learning of dependencies between sensor signals and tool wear states. Existing research indicates that multi-domain feature, encompassing the time domain, frequency domain, and time–frequency domain, effectively characterize tool wear while reducing signal redundancy [37]. In our experimental configuration, six-channel signals are collected via triaxial dynamometers (Fx, Fy, Fz) and triaxial accelerometers (Ax, Ay, Az). Consequently, extracting discriminative features from these signals is essential to precisely indicate tool wear conditions.

Multi-domain features are extracted from the preprocessed cutting signals of each channel. Detailed information on these multi-domain features is shown in Table 4. The time-domain features (Max, PP, Var, RMS, Ske, Kur) characterize the dynamic signal properties of tool wear through amplitude, energy, and distribution attributes; the frequency-domain features (SP, SS, SK) monitor harmonic component variations induced by wear based on spectral energy distribution. The time–frequency domain feature (WE) employs wavelet energy to capture transient impacts and gradual responses in non-stationary signals.

In Table 4, x_i represents the *i*-th data point of the collected cutting signal, where $i = 1, 2, \cdots, T$ and T is the total number of sampling points. $\bar{x}$ is the mean value of all the data in the cutting signal, f_i is the frequency spectrum, and $S(f_i)$ is the power spectral density. T_t represents the total number of samples used to compute the energy, and $d_{j,k}^{(M)}$ represents the wavelet packet coefficient at the *j*-th node and *k*-th frequency band in the specific wavelet decomposition level *M*.

Table 4
The extracted time-domain, frequency-domain, and time–frequency domain features.

Domain	Feature	Mathematical equation
Time	Maximum(Max)	$F_1 = max(x_i)$
	Peak-to-peak(PP)	$F_2 = max(x_i) - min(x_i)$
	Variance(Var)	$F_3 = \frac{1}{T}\sum_{i=1}^{T}(x_i - \bar{x})^2$
	Root mean square(RMS)	$F_4 = \sqrt{\frac{1}{T}\sum_{i=1}^{T}x_i^2}$
	Skewness(Ske)	$F_5 = \frac{1}{T}\sum_{i=1}^{T}\left(\frac{x_i - \bar{x}}{\sqrt{F_3}}\right)^3$
	Kurtosis(Kur)	$F_6 = \frac{1}{T}\sum_{i=1}^{T}\left(\frac{x_i - \bar{x}}{\sqrt{F_3}}\right)^4$
Frequency	Spectral power(SP)	$F_9 = \sum_{i=1}^{T}S(f_i)(f_i)^3$
	Spectral skewness(SS)	$F_8 = \sum_{i=1}^{T}\left(\frac{f_i - \mu}{\sigma}\right)^3 S(f_i)$
	Spectral kurtosis(SK)	$F_9 = \sum_{i=1}^{T}\left(\frac{f_i - \mu}{\sigma}\right)^4 S(f_i)$
Time-frequency	Wavelet energy(WE)	$F_{10} = \frac{1}{T_t}\sum_{i=1}^{T_t}\left(d_{j,k}^{(M)}\right)^2$

3.2.2. Feature selection using Spearman correlation coefficient (SCC)

Compared with multi-sensor signals, multi-domain features exhibit significantly reduced redundancy and dimensionality. However, not all features demonstrate substantial sensitivity to tool wear state. Consequently, feature selection becomes imperative to optimally preserve the most sensitive information while accurately capturing the relationship between the monitoring signals and the tool wear state. SCC is a non-parametric statistical measure that assesses the strength of association between two variables by analyzing their rank order. While it shares similarities with the PCC, yet it is better suited for evaluating the monotonicity and trend of the relationship between variables. As a result, SCC is widely used in feature selection tasks to capture the nonlinear relationships between features and target variables. The definition of SCC is as follows:

$$\rho_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)} \# \tag{13}$$

$$d_i = r(x_i) - r(y_i) \# \tag{14}$$

Where ρ_s is the SCC between features and target variables, with a range of $[-1, 1]$. When $\rho_s > 0$, it indicates a positive correlation; when $\rho_s < 0$, it indicates a negative correlation; and when $\rho_s = 0$, it indicates no correlation. n denotes the number of data samples, i.e., the number of cutting cycles; x_i denotes the feature vector of the i-th cutting process in the multi-domain feature set; y_i is the tool wear value corresponding to the i-th cutting process; $r(x_i)$ and $r(y_i)$ represent the ranks of the feature vector x_i and the tool wear value y_i in the i-th cutting process, respectively; d_i is used to measure the relative ranking difference between the feature vector x_i and the tool wear value y_i in the i-th cutting process, which is then used to calculate the SCC between features and target variables. $\sum_{i=1}^{n} d_i^2$ represents the sum of squared rank differences for all samples, which measures the sequence consistency between the feature vector x_i and the tool wear value y_i.

In TWM, SCC effectively captures the trend and relationship between features and tool wear values. Features with higher SCC values typically contain more wear-related information, which helps improve the accuracy and robustness of prediction models. By calculating SCC between each feature and the tool wear value, highly correlated sensitive features can be selected, avoiding redundant interference and significantly enhancing the efficiency and accuracy of TWM.

3.3. Data-driven model

3.3.1. BiLSTM-based initial model establishment

As an important extension of Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks enhance the ability to capture long-term dependencies by incorporating specialized gating structures. In this study, the force and acceleration signals acquired during the cutting process are represented as time-series data, which are particularly amenable to prediction using LSTM models. The architectural structure of an LSTM unit is illustrated in Fig. 5.

In Fig. 5, the input data X_t represents the sensitive feature set filtered by SCC; f_t is the forget gate, which regulates the quantity of information from the prior cell state that is transmitted to the current cell; i_t is the input gate, which determines the information that should be retained in the current cell state; and o_t is the output gate, which controls the quantity and type of information to be output.

The specific parameter update calculation process is shown in Eq. (15) to (20).

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \# \tag{15}$$

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \# \tag{16}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \# \tag{17}$$

$$C_t = f_t \odot C_{t-l} + i_t \odot \widetilde{C}_t \# \tag{18}$$

$$h_t = o_t \odot \tanh(C_t) \# \tag{19}$$

$$\widetilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \# \tag{20}$$

Where W_i, W_f, W_o and W_c are the weight matrices that measure the importance of the input in the input gate, the amount of information retained by the forget gate, the amount of aggregated input transferred to the hidden layer in the output gate, and adjusting the aggregated input in the input gate, respectively; b_i, b_f, b_o and b_c are the

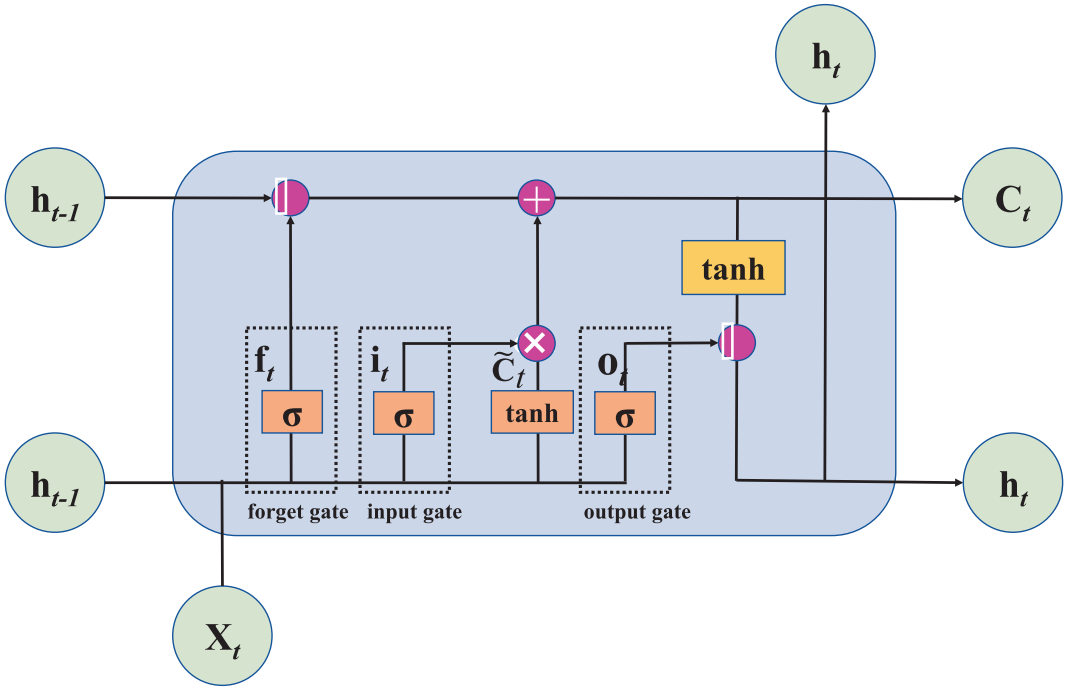


Fig. 5. The unit structure of LSTM.

corresponding biases; σ denotes the sigmoid activation function; tanh denotes the hyperbolic tangent activation function; $\odot$ denotes the element-wise multiplication; h_t is the memory, representing short-term memory; C_t and $\widetilde{C}_t$ denote the cell state and candidate state, respectively, with the latter awaiting consolidation into long-term storage.

However, standard LSTM networks have limitations in that they are incapable of propagating information from the backward direction to the forward direction and struggle to capture bidirectional dependencies effectively [38]. To overcome this limitation, the present study employs BiLSTM for tool wear prediction. The BiLSTM architecture employs two distinct LSTM layers: one processing the sequence forward and the other backward. This bidirectional design allows the model to integrate contextual information from both past and future states, overcoming the unidirectional constraint inherent in standard LSTMs. Consequently, BiLSTM effectively learns features from both temporal directions. The outputs of these layers are then concatenated to produce the final result. Fig. 6 visually compares the structures and highlights the differences between LSTM and BiLSTM. the forward hidden vector $\overrightarrow{h_i}$, the backward hidden vector $\overleftarrow{h_i}$, and the output h_i are shown in Eq. (21) to (23).

$$\overrightarrow{h_i} = \overrightarrow{LSTM}(X_t), i \in [1, n]\# \tag{21}$$

$$\overleftarrow{h_i} = \overleftarrow{LSTM}(X_t), i \in [1, n]\# \tag{22}$$

$$h_i = \left[\overrightarrow{h_i}, \overleftarrow{h_i}\right], i \in [1, n]\# \tag{23}$$

3.3.2. Multi-head attention mechanism-based dynamic weight allocation

The core function of the attention mechanism is to compute a weighted sum of input features, automatically identifying the features that are most critical for the tool wear prediction. This mechanism maps input features to three subspaces: Query (Q), Key (K), and Value (V). By calculating the dot product between the query and key, it assigns different weights to different features, reflecting their relevance in tool wear prediction. Key features receive higher weights, while less important features receive smaller weights. This correlation-based

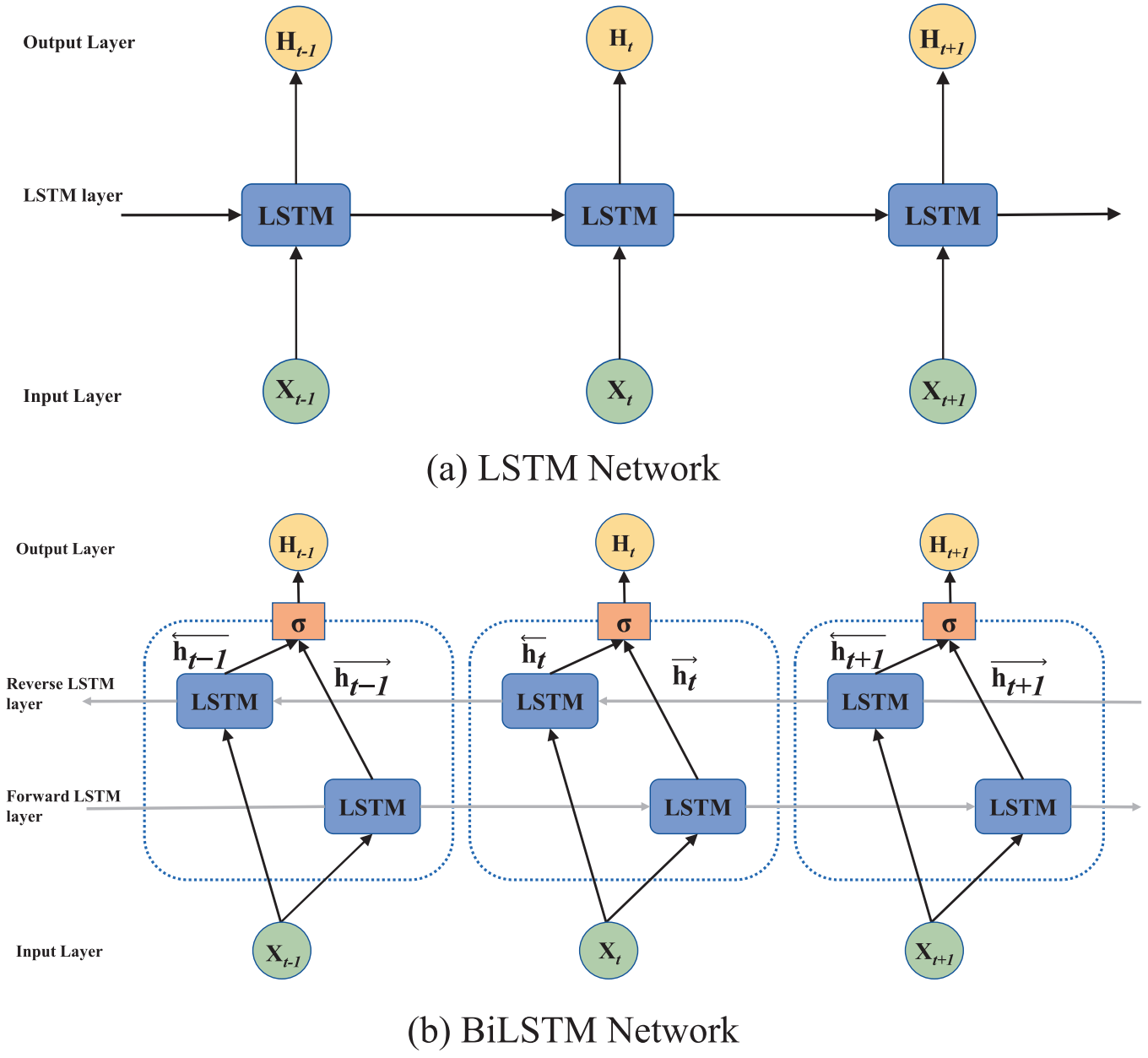


Fig. 6. LSTM and BiLSTM Networks.

weight allocation helps the model better capture the critical features in the tool wear process, improving prediction accuracy. The multi-head attention mechanism is shown in Fig. 7.

In this study, the input matrix X is the hidden state sequence generated by the BiLSTM model, consisting of hidden states $H_1, H_2, \ldots\ldots, H_n$ corresponding to each cutting operation. Through linear transformations, X is mapped to three different spaces: Q, K and V. This process is achieved by multiplying with weight matrices and can be represented by Eq. (24):

$$\begin{cases} Q = X^* W_Q \\ K = X^* W_K \# \\ V = X^* W_V \end{cases} \tag{24}$$

Where W_Q, W_K and W_V are the learnable linear transformation matrices for the Q, K and V, respectively.

In the attention mechanism, common calculation methods include additive attention [39] and dot-product attention [40]. Dot-product attention is especially well-suited for handling large-scale data in this task, owing to its rapid computation speed and minimal memory requirements. It can be formulated as follows:

$$A_i = softmax\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i, (i \in \{1, 2, \cdots, He\}, He = 8) \# \tag{25}$$

Where A_i denotes the output of the i-th attention head. He is the number of parallel attention heads. d_k is the dimensionality of the key vectors, used to scale the dot product results in order to prevent large dot product values from causing gradient vanishing or explosion. The *softmax* function normalizes the product of Q and K, thereby obtaining the weight distribution.

Unlike the single attention mechanism, the multi-head attention mechanism maps Q, K and V to multiple subspaces and simultaneously captures different information of the input features from multiple perspectives. Each head independently computes the attention for its subspace, and the final attention output is produced by concatenating the outputs from all attention heads and applying a linear transformation. The formula is expressed as:

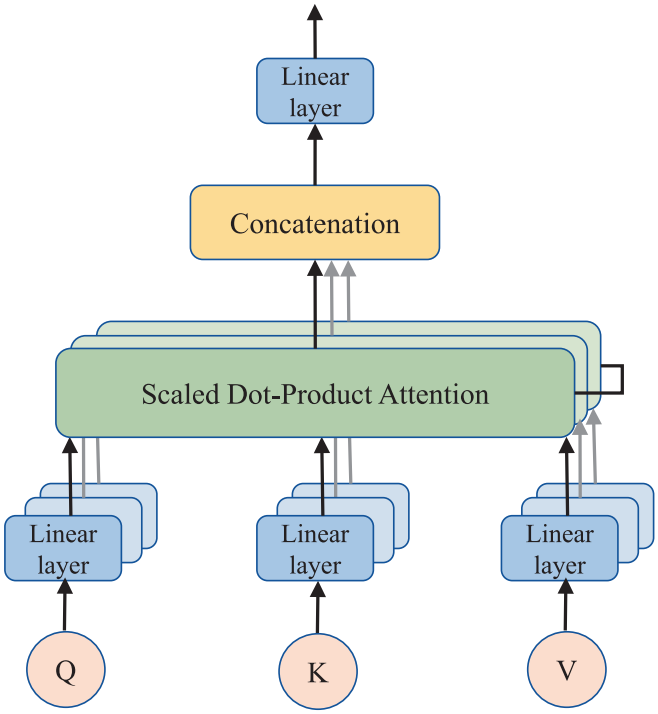


Fig. 7. The multi-head attention mechanism.

$$Attention(Q, K, V) = Concat(A_1, A_2, \cdots, A_{He}) W_O \# \tag{26}$$

Where $Attention(Q, K, V)$ represents the final attention output, and W_O is the learnable linear transformation matrix. Through this mechanism, the model can analyze time-series data from different perspectives, capturing complex patterns and dependencies in the tool wear process, which ultimately improves the overall prediction accuracy.

3.3.3. Integrated BiLSTM-AM predictive model construction

This study designs a data-driven model that combines BiLSTM and the attention mechanism (BiLSTM-AM). The BiLSTM-AM network structure is shown in Fig. 8.

Firstly, the model's input consists of multi-domain features X_1, $X_2 \cdots\cdots X_n$, which are the sensitive feature set filtered by SCC. These features contain information about the changes in tool wear after each cutting operation. Next, BiLSTM extracts both forward and backward features of the input sequence through the bidirectional LSTM layers, outputting the hidden states $H_1, H_2 \cdots\cdots H_n$ for each cutting operation. This helps capture the hidden information of the time series related to both past and future events.

The model incorporates multi-head attention mechanism to prioritize salient features. This mechanism projects input hidden states $H_1, H_2 \cdot \cdots\cdots H_n$ into distinct representation subspaces via independent linear mappings. Then, attention scores are computed independently within each subspace. Each attention head independently calculates attention and aggregates the results in this mechanism. The results from every head are concatenated and passed through a linear transformation to obtain the combined output representation H_d. This mechanism enables the model to focus on features from multiple perspectives at different cutting stages, dynamically adjusting its attention to the critical stages of the tool wear process, thus improving prediction accuracy.

Finally, the aggregated multi-head attention output H_d is fed into a fully connected layer to output the predicted tool wear value. This network combines the bidirectional dependency capability of BiLSTM in time-series modeling with the adaptive feature focusing ability of multi-head attention, significantly improving the accuracy of tool wear prediction.

3.4. Hybrid tool wear prediction model utilizing UKF

UKF is an improved version of the Kalman Filter (KF), specifically designed for state estimation in nonlinear systems. Unlike KF, which relies on linearization, UKF employs an unscented transformation to represent the statistical properties of the system state using a set of sigma points, effectively approximating nonlinear transformations. This approach eliminates the linearization assumptions inherent in system and observation models, allowing UKF to more accurately handle nonlinear dynamics. This is particularly advantageous when the state transition and observation equations are nonlinear. Tool wear, being a typical nonlinear process with complex characteristics that vary over time and under different cutting conditions, poses significant challenges for KF. In contrast, UKF, through its unscented transformation, offers a more precise estimation of tool wear, effectively adapting to this complex and dynamic process.

This study utilizes UKF to fuse the physics-based model with the data-driven model, achieving more accurate predictions. The physics-based model provides prior information about the tool wear process, capturing the overall trend of tool wear progression. Although it may involve certain simplifications and assumptions, it effectively represents the fundamental mechanisms of tool wear. In contrast, the data-driven model is trained on actual collected data, allowing it to flexibly reflect the tool wear state, though it is more sensitive to the quality of the data. Through the prediction and updating mechanism of UKF, this study dynamically integrates the strengths of both models while mitigating their respective weaknesses, resulting in a more accurate prediction of

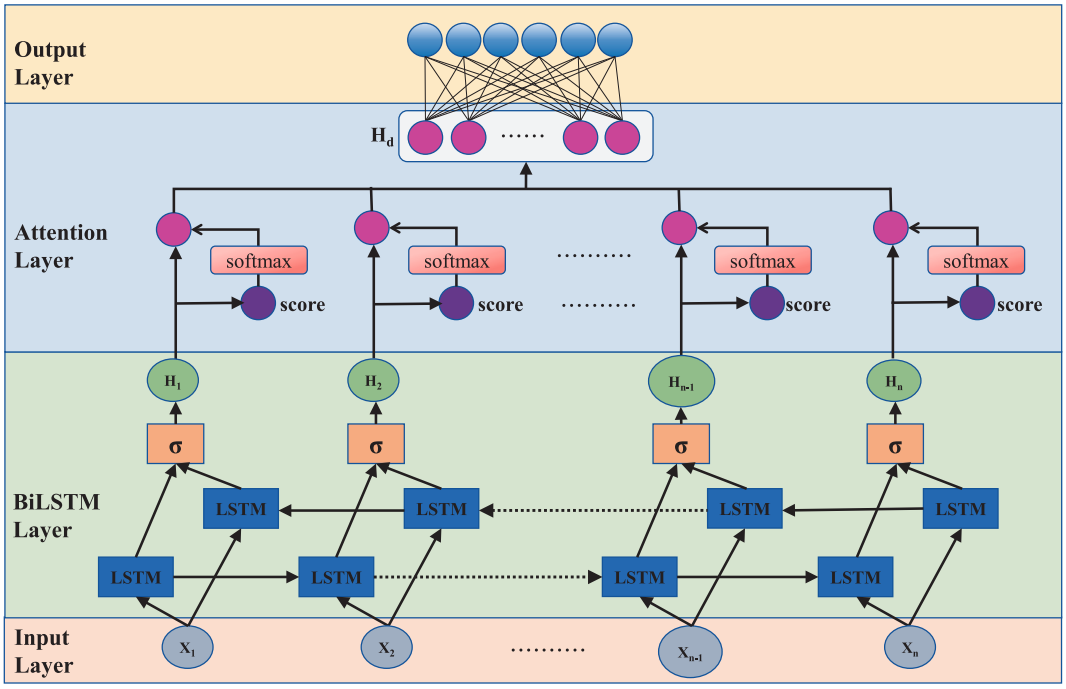


Fig. 8. BiLSTM-AM Network.

the tool wear state.

UKF is based on a state-space model to describe the dynamic characteristics of the system. Its core lies in dynamically estimating the tool wear state by combining the theoretical predictions from the physics-based model and the observations provided by the data-driven model. The UKF model mainly consists of the following two sets of equations:

The state fusion equation as:

$$X_{k+1} = X_k + \Delta f_{PM}(k+1) + \omega_k, \omega_k \sim N(0, Q_k)\# \tag{27}$$

Where X_{k+1} represents the tool wear state at the $(K+1)$-th moment, X_k represents the tool wear state at the K-th moment, $\Delta f_{PM}(k+1)$ is the increment of tool wear predicted by the physics-based model at the $(K+1)$-th moment, ω_k is the process noise, which reflects the uncertainty in the physics-based model, and Q_k is the process noise covariance matrix, reflecting the long-term statistical characteristics of the prediction error of the physics-based model.

The observation equation as:

$$Z_k = X_k + \upsilon_k, \upsilon_k \sim N(0, R_k)\# \tag{28}$$

Where Z_k is the observed value at the K-th moment, which corresponds to tool wear prediction $f_{DDM}(K)$ generated by the data-driven model, $f_{DDM}(K)$ is treated as a noisy observation of X_k, υ_k is the measurement noise, which reflects the random error in the prediction of the data-driven model, and R_k is the measurement noise covariance matrix, reflecting the long-term statistical characteristics of observational errors in the data-driven model.

In this study, the increment of tool wear predicted by the physics-based model, $\Delta f_{PM}(k+1)$, serves as the prior estimate of the tool wear state. The prediction from the data-driven model, $f_{DDM}(K)$, is then treated as the observed value to correct the uncertainty in the physics-based model's prediction. This hybrid approach not only bypasses the challenge of explicitly constructing state transition equations in the complex physics-based model but also compensates for random errors by incorporating the data-driven model, consequently improving prediction accuracy.

UKF handles the state estimation of nonlinear systems through the unscented transformation. It mainly consists of two stages: time update (prediction) and observation update (correction).

1. Time Update: The time update uses the prediction from the physics-based model to make a prior estimate of the tool wear state. The specific steps include the following

(1) Unscented Transformation: The unscented transform (UT) generates $2n+1$ sigma points to represent the nonlinear propagation of the tool wear state. Here, n denotes the dimension of the state vector. In this study, the state vector contains only the tool wear amount (i.e., $n=1$), resulting in three sigma points:

$$\mathscr{X}_k^i = \begin{cases} \widehat{x}_{k|k}, & i=0 \\ \widehat{x}_{k|k} + \sqrt{(n+\lambda)P_{k|k}}, & i=1 \\ \widehat{x}_{k|k} - \sqrt{(n+\lambda)P_{k|k}}, & i=2 \end{cases} \# \tag{29}$$

Where $\mathscr{X}_k^i$ are a set of sigma points used to represent the statistical distribution of the state variables, $\widehat{x}_{k|k}$ is the posterior state estimate at the K-th moment, $P_{k|k}$ is the posterior state covariance matrix at the K-th moment, and λ is a scaling factor, typically set as $\lambda = \alpha^2(n+g) - n$, where α and g are adjustment parameters used to control the distribution characteristics of the sigma points.

(2) State Propagation: Ideally, the physics-based model $f_{PM}(\cdot)$ is applied to each sigma point X_k^i to generate the state points at the next time step.

$$X_{k+1|k}^i = f_{PM}(X_k^i)\# \tag{30}$$

Due to the complexity of the physics-based model used in this study, it may not be feasible to directly apply the nonlinear transformation to each sigma point. In the state propagation stage, the increment of tool wear predicted by the physics-based model $\Delta f_{PM}(k+1)$ is combined with the uncertainty term ϵ^i to approximately describe the state propagation of each sigma point:

$$\mathscr{X}_{k+1|k}^i = \mathscr{X}_k^i + \Delta f_{PM}(k+1) + \epsilon^i, \epsilon^i \sim N(0, Q_k)\# \tag{31}$$

Where $X_{k+1|k}^i$ denotes the predicted state value of the i-th sigma point at the $(K+1)$-th moment, ϵ^i is the corresponding offset to each sigma point, used to reflect the uncertainty in the state propagation process.

(3) State Prediction: The weighted average of all transformed sigma points is carried out to obtain the prior estimation value of the tool wear state at the $K+1$-th moment.

$$\widehat{x}_{k+1|k} = \sum_{i=0}^{2n} W_i^{(m)} X_{k+1|k}^i \# \tag{32}$$

Where $\widehat{X}_{k+1|k}$ represents the prior estimate value of the tool wear state at the $K+1$-th moment.

$W_i^{(m)}$ is the weighting coefficient, typically set as:

$$W_i^{(m)} = \begin{cases} \dfrac{\lambda}{1+\lambda}, & i=0 \\ \dfrac{1}{2(1+\lambda)}, & i=1,2 \end{cases} \# \tag{33}$$

Through the weighted average, UKF avoids the linearization errors of KF and accurately captures the nonlinear dynamics of the system.

(4) Calculation of Prior Covariance:

$$P_{k+1|k} = \sum_{i=0}^{2n} W_i^{(c)} \left(X_{k+1|k}^i - \widehat{x}_{k+1|k}\right)^2 + Q_k \# \tag{34}$$

Where $P_{k+1|k}$ is the prior covariance matrix at the $(K+1)$-th moment.

$W_i^{(c)}$ is the weighting coefficient, typically set as:

$$W_i^{(c)} = \begin{cases} W_0^{(m)} + (1-\alpha^2+\beta), & i=0 \\ W_i^{(m)}, & i=1,2 \end{cases} \# \tag{35}$$

where β is a adjusting parameter;。.

2. Observation Update: The observation update corrects the prior estimate value of the tool wear state using the observed values to improve the accuracy of the state estimation. The main steps are as follows

(1) Compute the Kalman Gain:

$$K_{k+1} = P_{k+1|k}\left(P_{k+1|k} + R_{k+1}\right)^{-1} \# \tag{36}$$

Where K_{k+1} is the Kalman gain, which dynamically adjusts the weighting of the predictions of the physics-based model and the data-driven model, In the calculation of the Kalman gain, if $P_{k+1|k}$ is large, it indicates high uncertainty in the prediction of the physics-based model, and thus the weight of the observation value is higher. Conversely, if R_{k+1} is large, it indicates high observation error in the data-driven model, and the prediction of the physics-based model will dominate.

(2) Correct the State Estimate:

$$\widehat{x}_{k+1|k+1} = \widehat{x}_{k+1|k} + k_{k+1}\left(z_{k+1} - \widehat{x}_{k+1|k}\right) \# \tag{37}$$

Where $\widehat{x}_{k+1|k+1}$ is the posterior state estimate of the tool wear at the $K+1$-th moment, which is the corrected tool wear prediction, z_{k+1} is the observed value, which is the prediction at the $K+1$-th moment from the data-driven model, $\widehat{x}_{k+1|k+1}$ combines the trend information from the physics-based model with the real-time observations from the data-driven model, achieving a complementary fusion of both.

(3) Update the Covariance Estimate:

$$P_{k+1|k+1} = (1 - K_{k+1})P_{k+1|k} \# \tag{38}$$

Where $P_{k+1|k+1}$ is the posterior covariance matrix at the $K+1$-th moment, representing the updated estimation of uncertainty. Through this update, UKF can adapt to the changing system uncertainties while continuously performing state estimation, providing a more accurate tool wear state estimate.

UKF operates in a prediction-correction loop, as shown in Fig. 9. In the process of fusing the physics-based model and the data-driven model, UKF dynamically balances the strengths and weaknesses of both, achieving the accurate prediction of the tool wear state.

4. Experiment design

4.1. Experimental setup

The milling experiments were conducted on CNC machine to evaluate the effectiveness of the newly developed integrated approach for predicting tool wear. The tool selected was an uncoated flat-end mill made of aluminum-doped high-speed steel, with 4 teeth, a diameter of 8 mm, a rake angle of 14°, a clearance angle of 14°, and a helix angle of 35°. The workpiece material was mold steel with dimensions of 30 × 45 × 50 (mm). The data acquisition process is shown in Fig. 10. Under stable milling conditions, multi-sensor signals were gathered with no occurrence of resonance.

During the machining process, cutting forces acting along three perpendicular axes were measured utilizing a Kistler 9347C dynamometer, which was installed between the workpiece and the machine bed. Simultaneously, vibration signals were acquired via a Kistler 8763B accelerometer attached to the workpiece side. Both signals were conditioned using a charge amplifier and sampled continuously at a

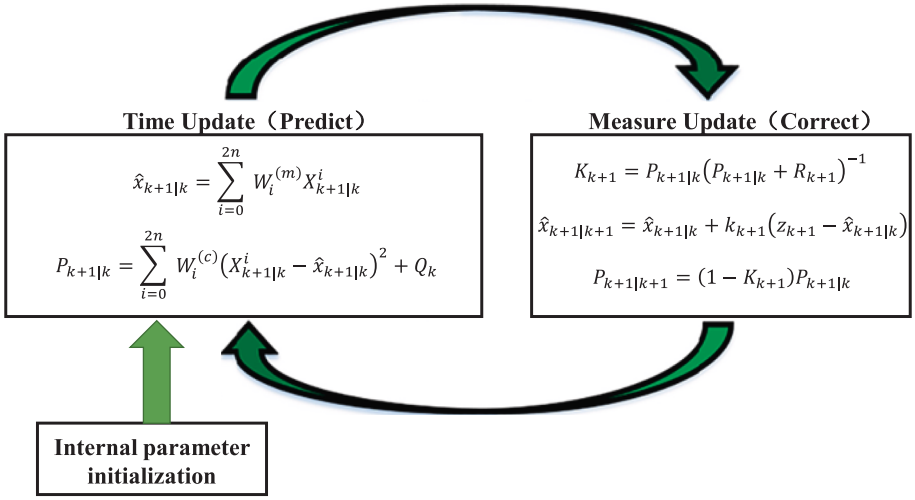


Fig. 9. Unscented Kalman Filter (UKF).

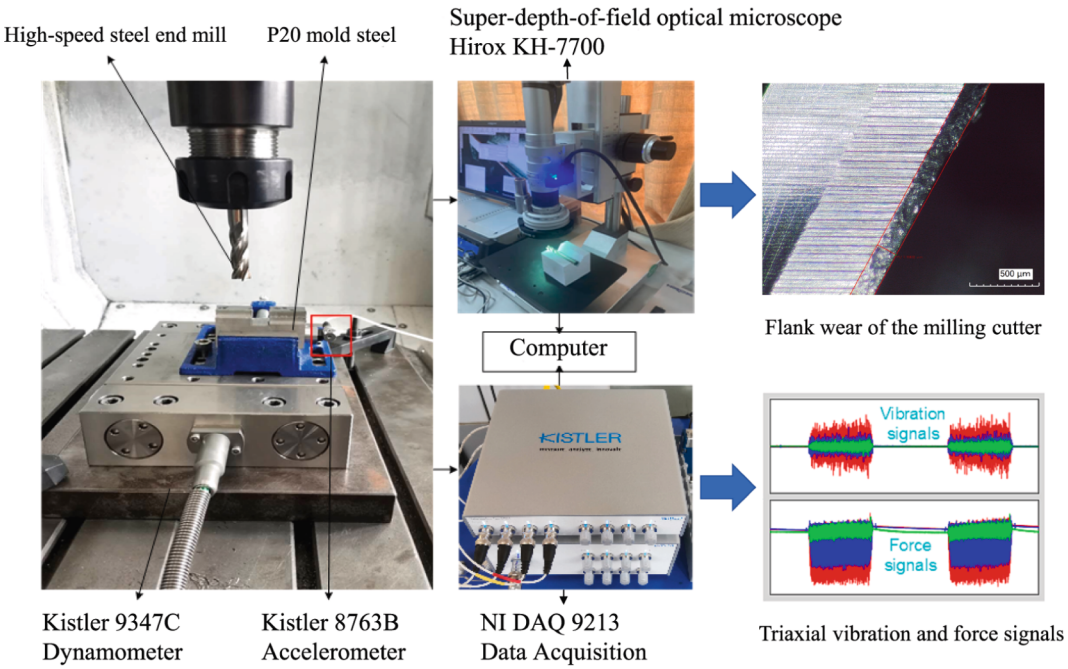


Fig. 10. The schematic diagram of the data acquisition process.

frequency of 2500 Hz through an NI-DAQ system for tool wear prediction. After each milling operation, flank wear occurring on each of the four cutting edges was measured with a Hirox KH-7700 optical microscope. The average value across these edges served as the definitive tool wear metric.

Three milling experiment sets used distinct machining parameters. While radial cutting depth remained constant, spindle speed, feed rate and axial cutting depth varied between sets. Table 5 details these parameters. Each set of milling experiments used 15 milling tools. For more details on the experimental setup and data acquisition, readers can refer to previous studies [41].

After completing the three sets of milling experiments, flank wear progression curves were obtained as shown in Fig. 11. Wear increased steadily with cutting time until tool failure occurred. This trend aligned with theoretical predictions, confirming the experimental validity. Therefore, the direct measurements of the flank wear obtained in the experiments can be used to validate the tool wear values predicted by the proposed method.

4.2. Data processing and analysis

The multi-sensor data acquisition was maintained continuously throughout each milling operation to avoid being disturbed by the processing process. Using the first dataset as an example, Fig. 12(a) reveals raw signals containing both cutting and non-cutting segments. Isolating the cutting phase was necessary to remove non-processing interference. However, the extracted signals contain outliers due to sporadic machining disturbances as shown in Fig. 12(b). Next, Hampel filtering is applied to replace data points exceeding three times the standard deviation with the average of the neighboring 1000 points, effectively removing the outliers from the cutting signals. The signal after outlier removal is shown in Fig. 12(c). Finally, wavelet packet decomposition is used to denoise the signal, with an 8-layer decomposition using the "db8″ wavelet base. The soft threshold method, based on the median absolute deviation (MAD), is applied for denoising, followed by signal reconstruction. The denoised signal is shown in Fig. 12(d).

Following multi-sensor data preprocessing, Table 4 details the extracted multi-domain features. These were normalized to enhance generalization. Fig. 13 displays normalized values for the first dataset, where Ax, Ay, Az and Fx, Fy, Fz denote acceleration and force components along x/y/z axes, respectively.

Fig. 13 illustrates that multi-domain features derived from multi-sensor signals evolve differently with increasing milling cycles. Certain features, including maximum value, peak-to-peak value, and variance, exhibit trends similar to tool wear. While others, such as spectral skewness in the Ay and Az directions, exhibit an opposite evolution trend. Additionally, some multi-domain features show limited sensitivity to changes in tool wear. Therefore, feature selection is crucial, as it helps filter out the most relevant and sensitive information from the signals that is highly associated with the tool flank wear.

SCC was then employed to quantitatively analyze the multi-domain features. The monotonicity of these features across the six channels (Fx, Fy, Fz, Ax, Ay, Az) is displayed in Fig. 14(a). Fig. 14(b) details the feature correlation coefficients, visually complementing the quantitative analysis. These results guide the subsequent selection of sensitive features.

As shown in Fig. 15(a), features exceeding an SCC threshold of 0.97 were identified to form the sensitive feature set. Furthermore, the monotonic trend of these selected features is illustrated in Fig. 15(b). It is

Table 5
Machining parameters of the milling operation.

Group	Cutter	Spindle speed (r/min)	Feed speed (mm/min)	Feed engagement (mm/r)	Axial cutting depth (mm)	Radial cutting depth (mm)	Material removal rate (mm^3/min)
1	No.01–15	1500	750	0.125	0.1	6	7.5
2	No.16–30	1500	375	0.0625	0.2	6	7.5
3	No.31–45	2000	500	0.0625	0.15	6	7.5

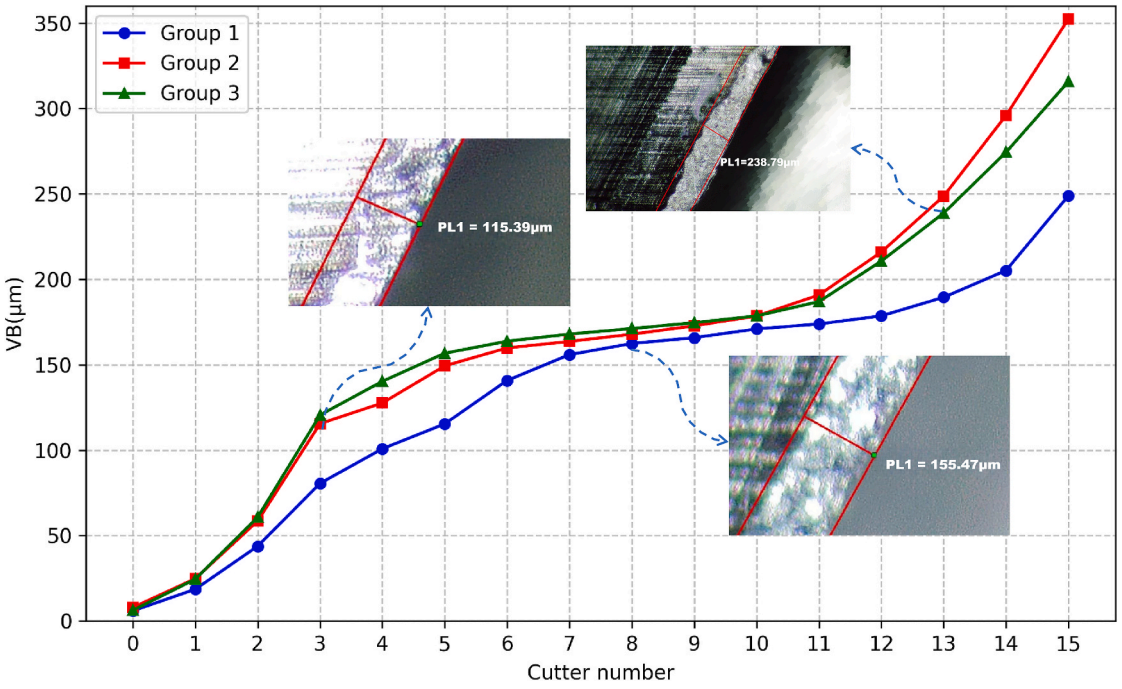


Fig. 11. The actual flank wear curves of the tree sets of milling tools and examples.

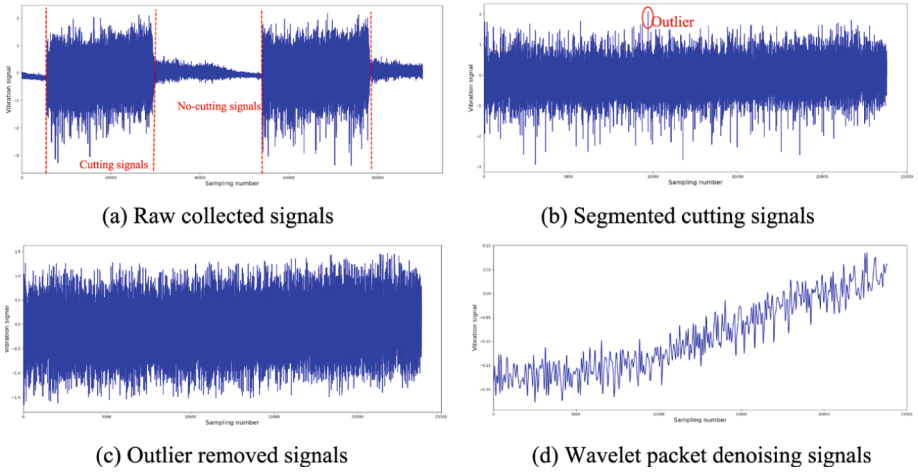


Fig. 12. An example of data preprocessing for the acceleration signal in the x-direction.

clear that the included features demonstrate clear trend characteristics, confirming that tool wear progressively increases with machining time. Therefore, the constructed sensitive feature set provides valuable and reliable information for TWM, forming a solid foundation for enhancing the accuracy and robustness of the data-driven model.

5. Results analysis and discussion

5.1. Tool wear prediction based on physics-based model

The iterative calculation of the analytical model for the flank wear of end mill, established in Section 3.1, was performed. And the predictions for the entire tool wear process were obtained as shown in Fig. 16. Under the working conditions for machining P20 die steel using aluminum-doped high-speed steel cutting tools, the machining temperature remains relatively low, limiting the mutual solubility of the materials. As a result, diffusion wear is not pronounced. However, the VB predictions of tool flank wear exhibit strong consistency with the actual flank wear values in terms of variation trends, thereby validating the accuracy and generalization capability of the adopted physics-based model.

In order to assess the predictive accuracy of the analytical model in a quantitative manner, RMSE, MAE, and R^2 were chosen as assessment indicators, with the results shown in Table 6. These results validate the accuracy as well as the generalization capability of the analytical model for the flank wear of end mill. However, due to the limitations of the physics-based model, there is still a significant discrepancy between the predicted results and experimental data, which may be caused by the following factors:

(1) The developed temperature field model of flank wear contains certain inaccuracies, which likely reduce the predictive accuracy of the analytical model.
(2) Vibrational influences were not incorporated into the analytical model, which may contribute to the discrepancies.
(3) The precision of the normal stress model of flank wear is relatively low, which further limits the model's precision.

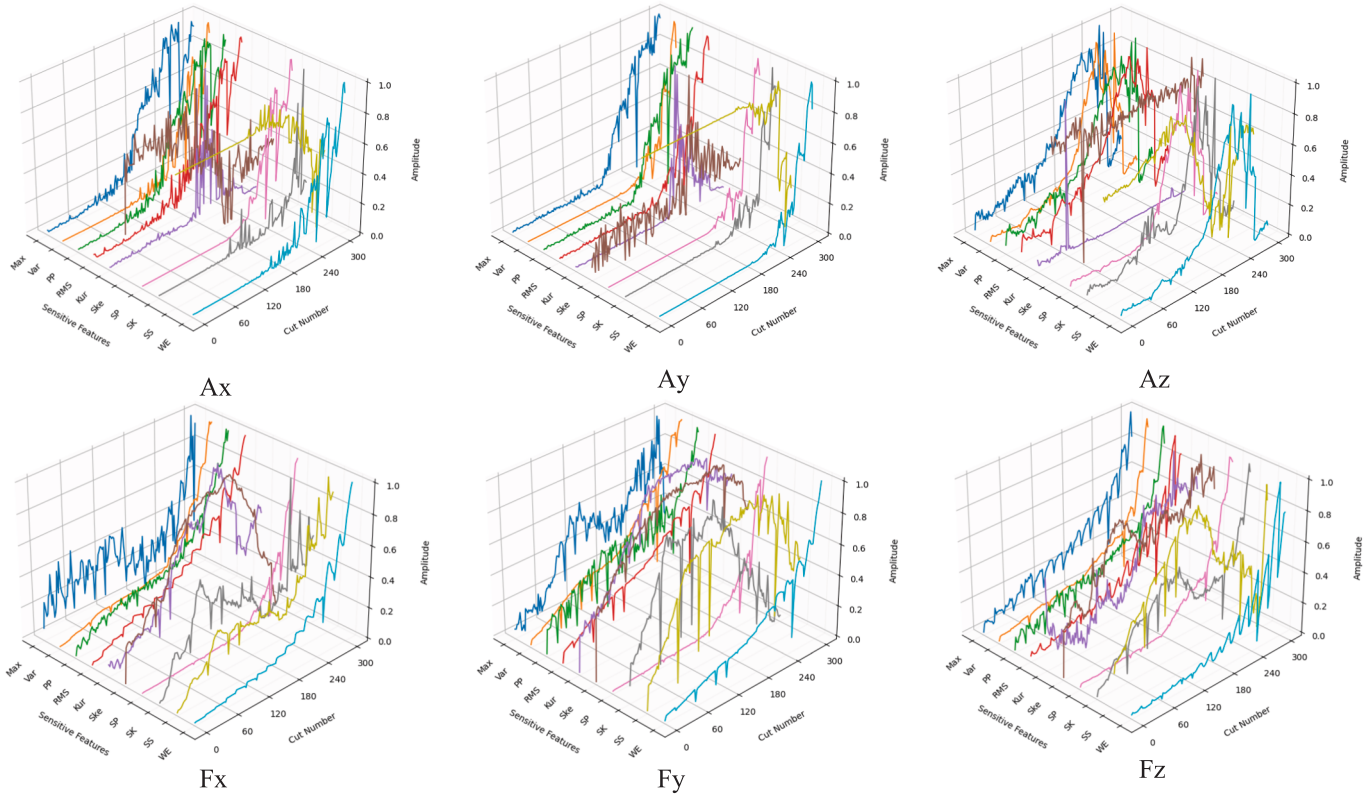


Fig. 13. The normalized multi-domain features of the first set of multi-sensor signals.

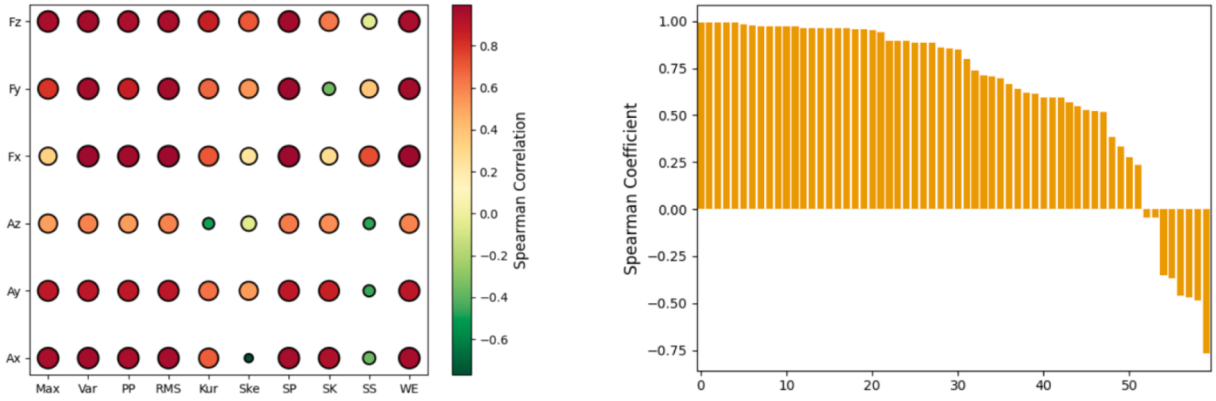


Fig. 14. Analysis of the extracted features.

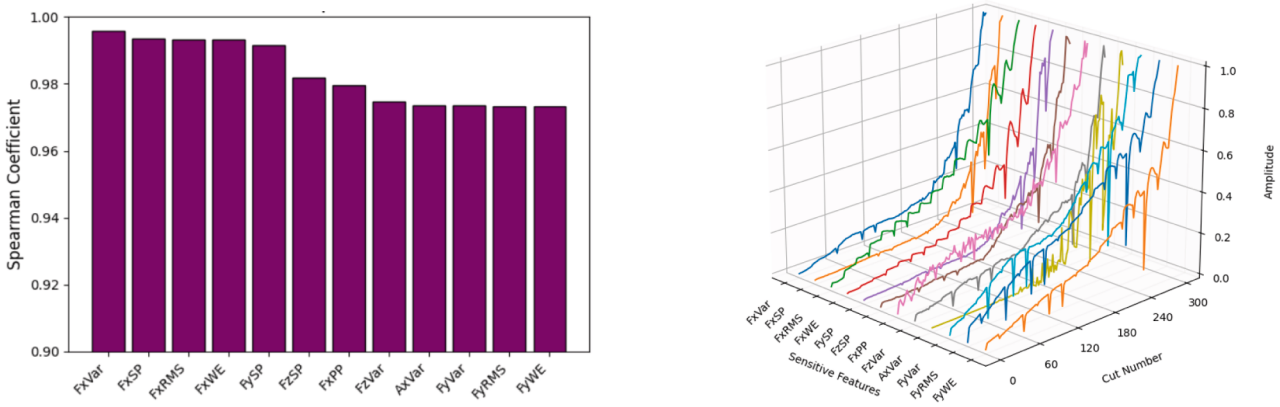
(4) Oxidation wear, which occurs on the flank during milling, was not accounted for the physics-based model. These factors are likely the primary causes of the observed deviations between the predicted and experimental data.

5.2. Multi-domain feature selection analysis

To evaluate the efficacy and advantages of SCC, a comparative study was carried out alongside two conventional feature selection and dimensionality reduction methods: PCC and Principal Component Analysis (PCA). Subsequently, these three methods were applied to three data-driven models: RNN, GRU, and LSTM.

PCC chooses features that have a correlation coefficient exceeding 0.9 according to the linear relationship between the features and the degree of tool wear. However, it is limited to capturing only linear relationships and may overlook potential nonlinear connections. PCA reduces the dimensionality by retaining the principal components that account for 95% of the variance. While this method helps to reduce dimensionality and preserve most of the information, it is constrained to linear data, and the resulting principal components are often difficult to interpret, which may lead to the loss of crucial features. In contrast, SCC evaluates the nonlinear correlation between features and tool wear values. By selecting sensitive features based on these correlations, SCC not only reduces the complexity of the data but also maximizes the retention of key features that are strongly correlated with tool wear. This approach ultimately enhances the accuracy of the prediction model.

This study uses RMSE, MAE, and R^2 as evaluation metrics to evaluate the effectiveness of the three methods in a quantitative manner across



(a) Selected sensitive features (b) Normalized sensitive feature curves

Fig. 15. Analysis of sensitive features.

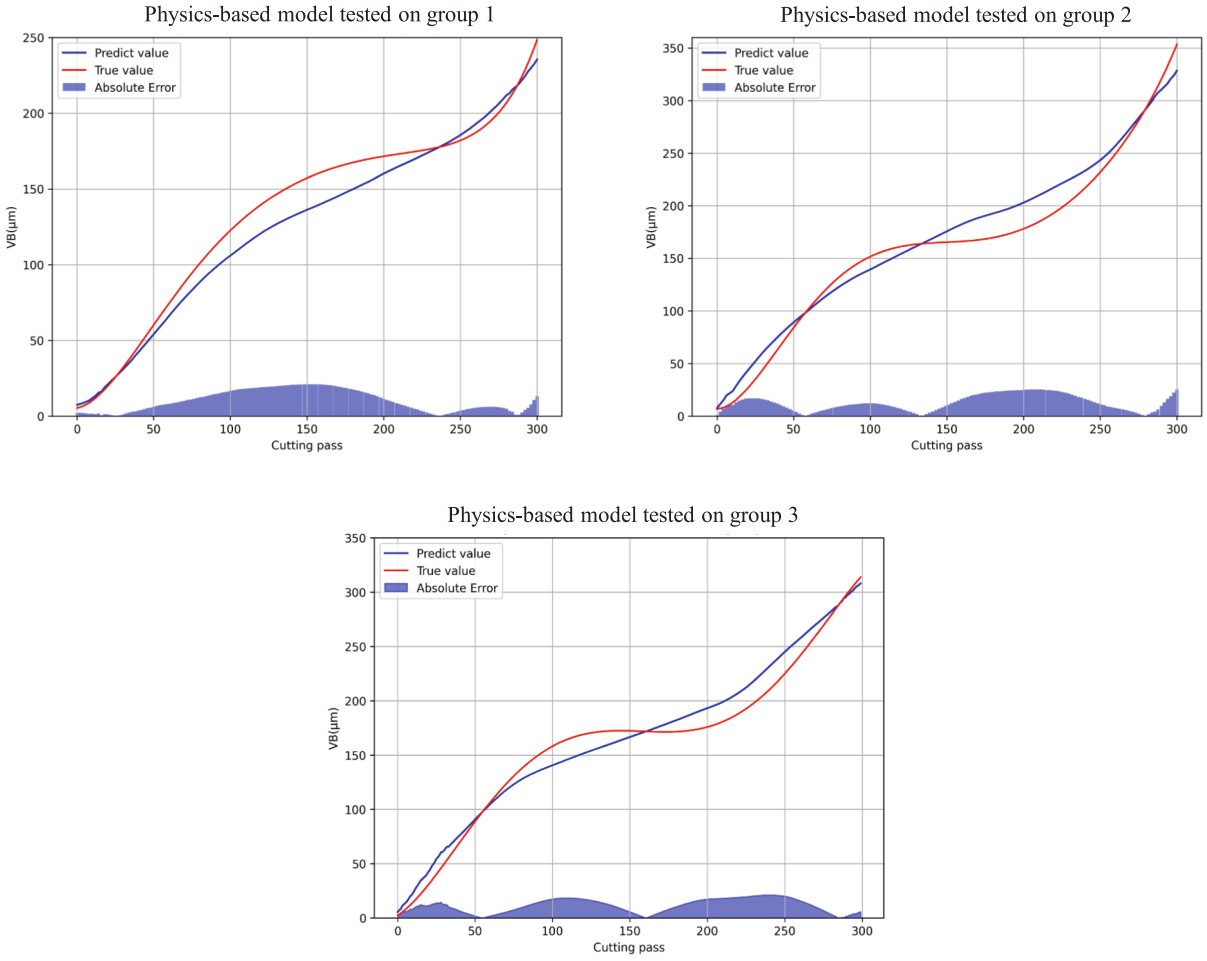


Fig. 16. Prediction results of tool wear physics-based model.

Table 6
Performance evaluation of physics-based model on three datasets.

Test samples	RMSE	MAE	R^2
Group 1	17.2837	14.0429	0.9171
Group 2	21.3474	18.3434	0.9233
Group 3	19.6126	17.1232	0.9279

different models. Table 7 and Fig. 17 show the performance of each method in different models, providing a basis for a comprehensive comparison of their effectiveness and superiority.

From Table 7 and Fig. 17, it is evident that SCC demonstrates better prediction accuracy for the other methods and achieves the best performance in the LSTM, further proving the superiority of SCC in tool wear prediction.

Table 7
Performance evaluation of multi-domain feature selection under different criteria.

Models	Schemes	RMSE	MAE	R^2
RNN	PCA	22.5435	19.8675	0.8091
	Pearson	16.5277	14.1544	0.8241
	Spearman	15.6255	12.2587	0.8418
GRU	PCA	20.9747	18.5098	0.8238
	Pearson	16.2257	13.9612	0.8719
	Spearman	14.5331	10.8732	0.8955
LSTM	PCA	20.9543	18.0253	0.8332
	Pearson	15.8456	12.1456	0.8891
	Spearman	13.4385	9.8735	0.9072

5.3. Tool wear prediction based on data-driven model

Through the performance evaluation and comparison of the different multi-domain feature selection and dimensionality reduction methods discussed above, the effectiveness and superiority of SCC was validated. Subsequently, it was then applied to the proposed data-driven model using BiLSTM-AM. This model was compared with BiGRU, BiLSTM, and BiGRU-AM to demonstrate its validity and advantages in predicting the tool flank wear.

To verify the effectiveness of the recognition performance of the proposed approach across different datasets, this study applied two sets of milling data to test four models. To effectively reduce the influence of data distribution randomness, overfitting, and test set information leakage during the training process, a sequential splitting strategy was implemented. The initial 80% of the dataset was designated as the training set, and the remaining portion, accounting for 20%, was utilized as the test set to assess the model's performance. The flank wear predictions and absolute errors are shown in Fig. 18. As illustrated in Fig. 18, it is evident that the predictions fluctuate near the actual tool wear values, with the BiLSTM-AM model exhibiting the smallest fluctuation range.

To facilitate the observation and analysis of the predictions and performance of each model, metrics statistics of different data-driven models are presented in Fig. 19 and Table 8. It is evident that BiLSTM-AM achieved the smallest MAE and RMSE, and the highest R2, outperforming all the comparison models. This indicates that, compared to

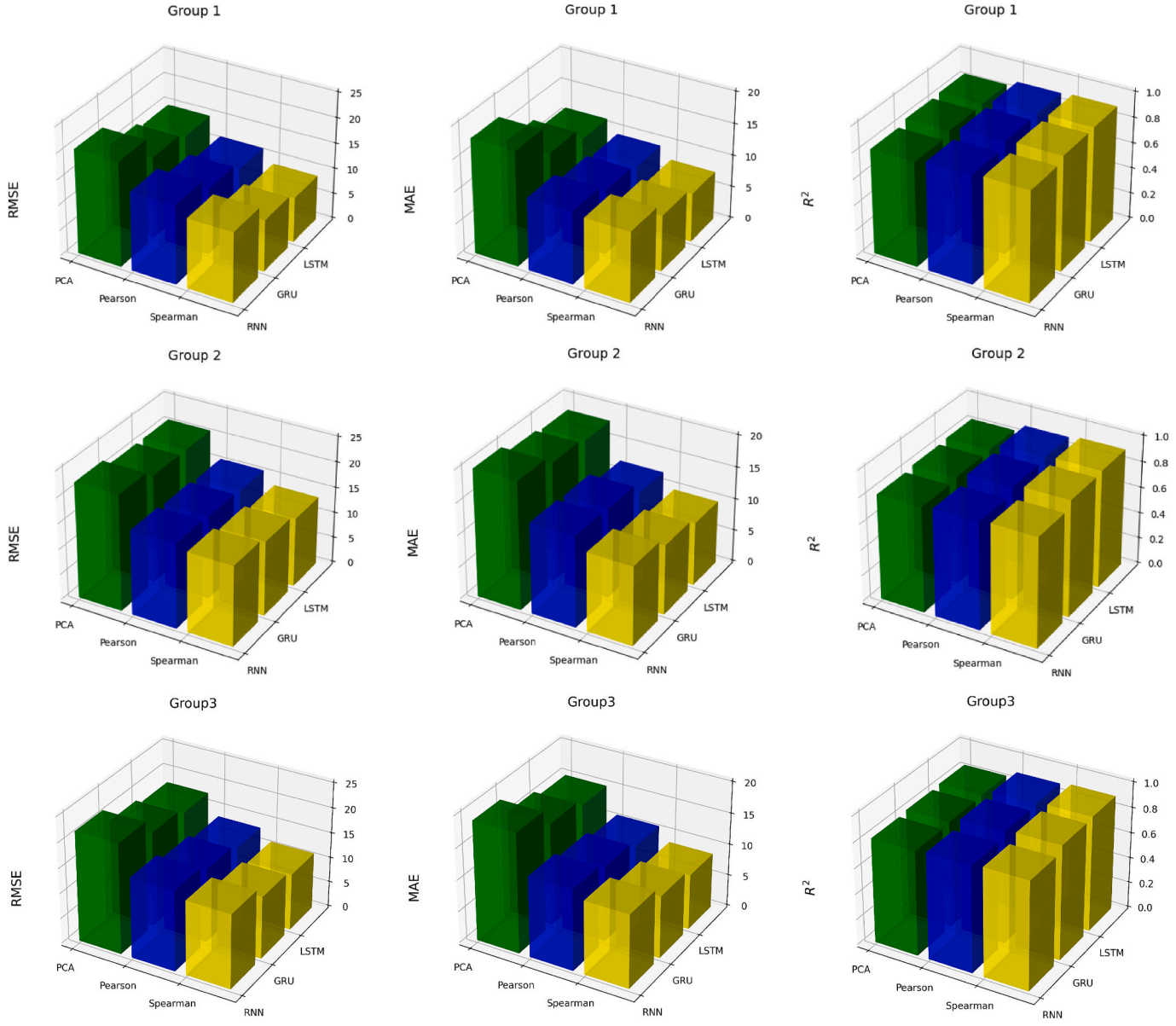


Fig. 17. Performance comparison of different feature selection.

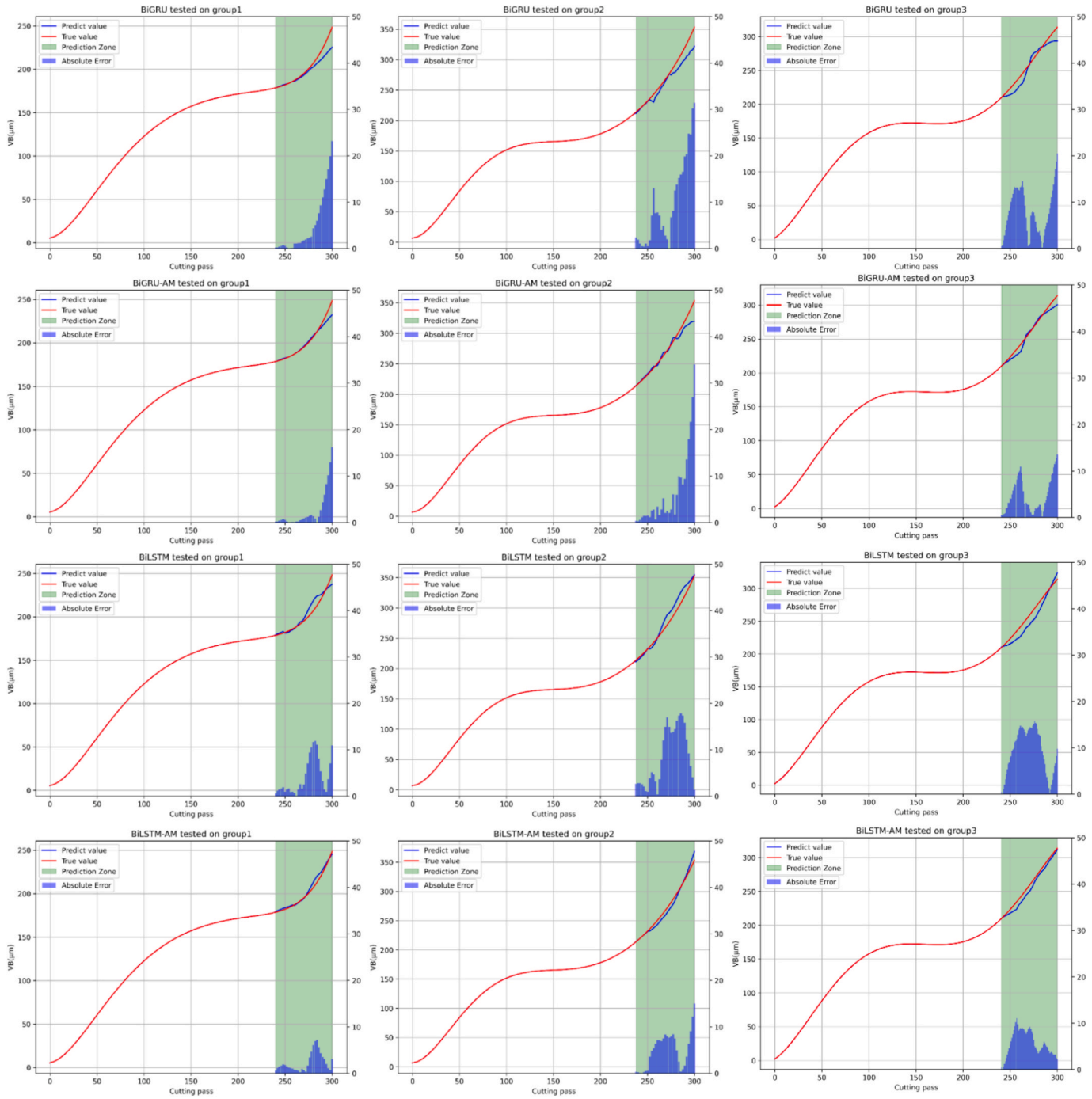


Fig. 18. Prediction Results of Data-Driven Models.

other models, the BiLSTM-AM model's tool wear predictions are closer to the actual measured values. These results validate the effectiveness and superiority of the proposed method in predicting the tool flank wear.

Overall, BiLSTM-AM achieved the smallest MAE and RMSE values, and the highest R^2 among all the comparison models. This superior performance can be attributed to BiLSTM 's stronger ability to model temporal dependencies and adapt to complex features compared to other models. Additionally, with the integration of the multi-head attention mechanism, the model can analyze time-series data from various perspectives, effectively capturing complex patterns and dependencies in the tool wear process, which improves the overall prediction accuracy. Therefore, the proposed method not only enhances the accuracy of the tool flank wear prediction but also increases its robustness across different milling processes.

5.4. Tool wear prediction based on hybrid model

To address the limitations of single prediction models and the current state of data-mechanism fusion research, which mostly focuses on a single mechanism or specific mechanism fusion, this study proposes a hybrid model based on UKF. The approach integrates a data-driven model with an analytical model to predict end mill flank wear, taking into account the combined effects of abrasive, adhesive, and diffusive wear mechanisms. Fig. 20 illustrates the comparison between the predictive outcomes of the individual model and the combined model.

From Fig. 20, it is clear that integrating the physics-based model with

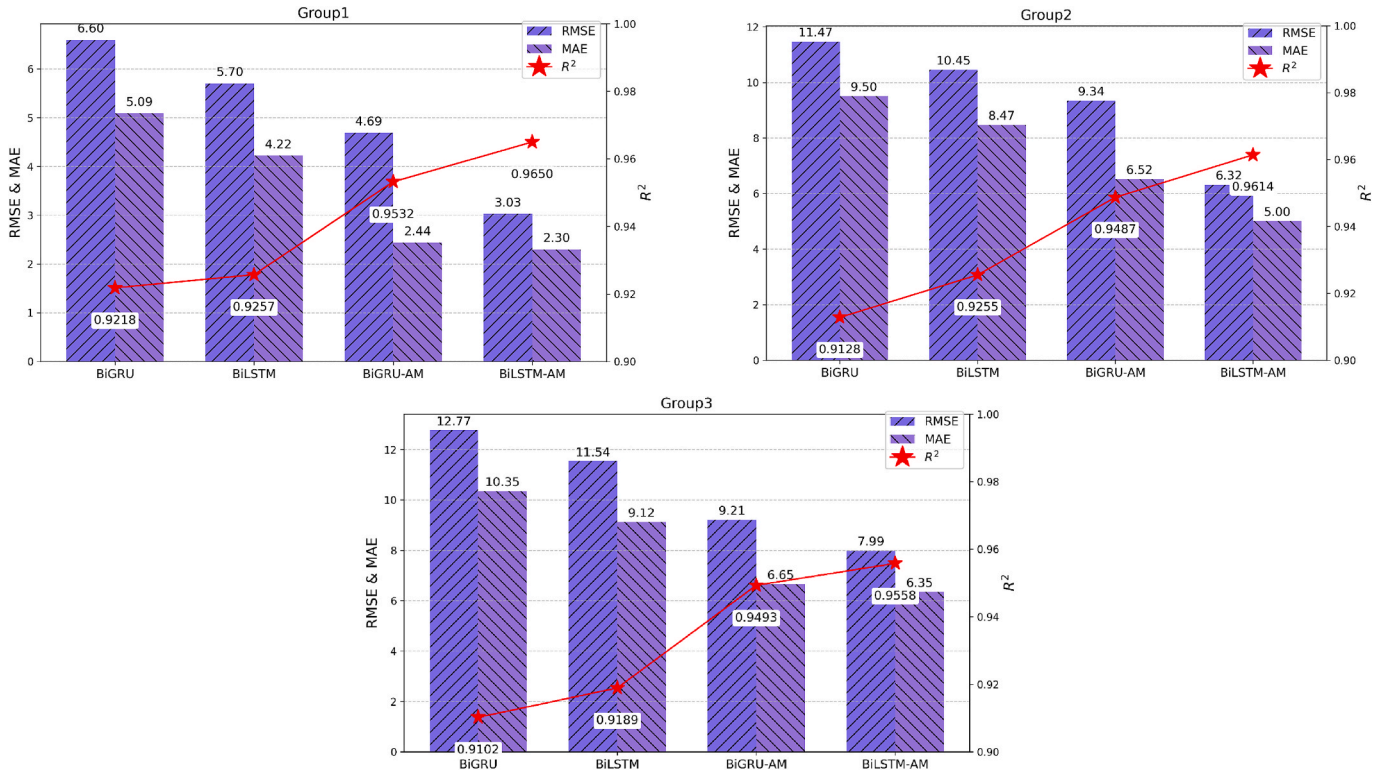


Fig. 19. Metrics statistics of different data-driven models.

Table 8
Performance evaluation of different data-driven models on three datasets.

Test samples	Schemes	RMSE	MAE	R^2
Group 1	BiGRU	6.5954	5.0882	0.9218
	BiLSTM	5.7010	4.2232	0.9257
	BiGRU-AM	4.6909	2.4390	0.9532
	BiLSTM-AM	3.0260	2.2985	0.9650
Group 2	BiGRU	11.4671	9.5049	0.9128
	BiLSTM	10.4499	8.4701	0.9255
	BiGRU-AM	9.3393	6.5192	0.9487
	BiLSTM-AM	6.3183	5.0040	0.9614
Group 3	BiGRU	12.7654	10.3456	0.9102
	BiLSTM	11.5432	9.1234	0.9189
	BiGRU-AM	9.2109	6.6543	0.9493
	BiLSTM-AM	7.9876	6.3456	0.9558

the data-driven model through the use of UKF leads to a noticeable outcome. The fluctuations in the predictions are significantly reduced, and the predictions align more closely with the actual wear values. To quantitatively evaluate the prediction performance of the three models, RMSE, MAE, and R^2 were chosen as evaluation metrics, with results shown in Table 9.

As seen from Table 9, when either the physics-based model or the data-driven model is used independently for predictions, the error between the experimental results and the predictions is relatively large. However, when the proposed hybrid method for tool wear prediction is applied, the prediction error is significantly reduced, with the predicted results closely aligning with the actual measured values. Compared to the predictions made by the single models, the hybrid method reduced the average RMSE by 46.5%, 80.45% and 55.11%, the average MAE by 46.75%, 80.90% and 57.22%, and increased R^2 to 99.15%, 99.31% and 98.85% for the three datasets, respectively. These improvements in prediction accuracy demonstrate the effectiveness and superiority of the proposed UKF-based hybrid method in tool wear prediction.

This verifies that the proposed method, combining the physics-based model with the data-driven model using UKF, significantly enhances the precision of tool wear predictions, offering substantial improvements over individual models.

5.5. Ablation experiments and results analysis

To verify the effectiveness of the proposed hybrid tool wear prediction model based on UKF, this study set up ablation experiments for comparative analysis. As shown in Table 10, this hybrid model is significantly superior to the other two comparative models in terms of RMSE, MAE, and R^2 indicators. One is the data-driven model that only uses BiLSTM-AM, and the other is the static hybrid model that linearly weights and averages the prediction results of the physics-based model and BiLSTM-AM with fixed weights. It should be noted that the fusion strategy of the static hybrid model is $VB_{fused} = W_s{*}VB_{DDM} + (1 - W_s){*}VB_{PM}$, where VB_{fused} is the fusion result, VB_{DDM} is the prediction results of the data-driven model, and VB_{PM} is the prediction results of the physics-based model. The weight W_s is determined through the validation set test. In this experiment, candidate values such as 0.3, 0.5, and 0.7 were tested, and finally $W_s = 0.7$ with the optimal performance was selected as the comparison benchmark.

As shown in Table 10, the prediction stability of pure data-driven models is insufficient. Although the linear static fusion can reduce errors to a certain extent, its fixed weight strategy is difficult to adapt to the dynamic changes in the wear process. In contrast, the hybrid model for tool wear prediction based on the UKF can adaptively achieve deep fusion by real-time adjusting the fusion weights of the physics-based model and the data-driven model. Specifically, it relies on physical laws during the stage of stable trends and places more emphasis on data-driven methods during the severe wear stage with intense non-linearity. This is the fundamental reason for the significant improvement in its performance. These results fully demonstrate the core contribution of the UKF dynamic fusion mechanism.

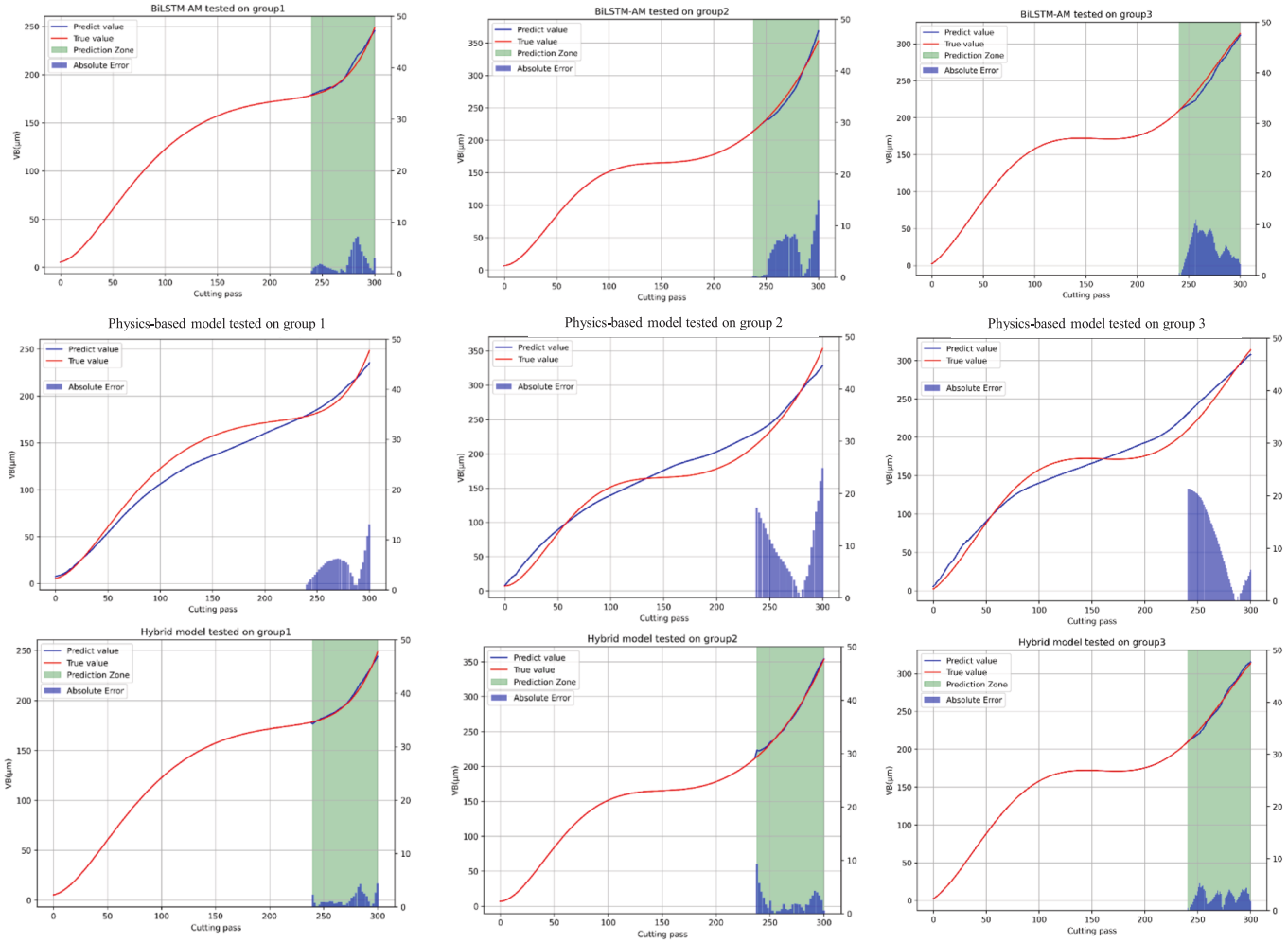


Fig. 20. Prediction results of different tool wear prediction models.

Table 9
Performance evaluation of different tool wear prediction models on three datasets.

Test samples	Schemes	RMSE	MAE	R^2
Group 1	BiLSTM-AM	3.0260	2.2985	0.9650
	physics-based model	11.9937	8.5829	0.8813
	Hybrid model	1.8655	1.4536	0.9915
Group 2	BiLSTM-AM	6.3183	5.0040	0.9614
	physics-based model	10.3774	8.0434	0.8960
	Hybrid model	2.8582	2.1672	0.9931
Group 3	BiLSTM-AM	7.9876	6.3456	0.9558
	physics-based model	13.4926	10.3932	0.8795
	Hybrid model	4.3210	3.2567	0.9885

Table 10
Performance comparison of UKF dynamic fusion in ablation experiments.

Test samples	Schemes	RMSE	MAE	R^2
Group 1	BiLSTM-AM	3.0260	2.2985	0.9650
	Static Fusion	2.5780	1.9723	0.9752
	Hybrid model	**1.8655**	**1.4536**	**0.9915**
Group 2	BiLSTM-AM	6.3183	5.0040	0.9614
	Static Fusion	5.1267	3.9815	0.9768
	Hybrid model	**2.8582**	**2.1672**	**0.9931**
Group 3	BiLSTM-AM	7.9876	6.3456	0.9558
	Static Fusion	6.6417	5.2478	0.9689
	Hybrid model	**4.3210**	**3.2567**	**0.9885**

5.6. Discussion

5.6.1. Reliability discussion of physics-based models

It should be noted that there are huge challenges in directly and accurately measuring the transient temperature and stress at the interface between the tool and the workpiece. Therefore, the sub-models (including the temperature field model, stress model, and various flank wear model) adopted in the physics-based model constructed in this study are all derived from classic literatures [27,31,32,34] that has been widely verified in this field. These models have been proven to be able to effectively reflect the key physical phenomena during the cutting process through high-fidelity simulation or indirect experimental methods. Although the intermediate variables have not been independently measured, as shown in Fig. 16, the prediction trend of the final tool wear amount by this integrated physics-based model is consistent with the experimental measurement values, which proves the rationality and effectiveness of its description of the tool wear process from a systematic perspective.

5.6.2. Discussion on the generalization ability of the model

The proposed hybrid model demonstrates excellent predictive performance under specific cutting conditions. However, regarding its wide application in industry, a natural question is the generalization ability and migration cost of the model. This study candidly points out that when there are fundamental changes in the application scenario, such as machining a completely new material (from steel to titanium alloy) or using a cutting tool with completely different geometric parameters, the

model needs to be adjusted specifically. This process involves two aspects:

- Adjustment of the physics-based model: The material-related parameters in the model (such as the mechanical and thermal parameters of the workpiece material) and the tool-related parameters (such as the geometric parameters of the tool and the tool material parameters) need to be updated to the values corresponding to the new working conditions. These parameters can usually be obtained through material handbooks, tool manufacturer catalogs, or simple standard tests.
- Update of the data-driven model: To adapt to new working conditions, model migration needs to complete data preparation, feature reconstruction, and model update in sequence. First, collect new data and perform preprocessing, and then extract and screen features to construct health indicators. Subsequently, use the new data to retrain or fine-tune the model. Finally, verify the model accuracy through actual wear data and iterate for optimization.

However, the advantage of this hybrid model lies in its dynamic fusion mechanism based on the UKF. This mechanism can automatically balance the contributions of physical laws and data characteristics under new working conditions by real-time adjustment of the confidence levels in them. It simultaneously takes advantage of the trend inference ability of the physics-based model and the advantage of the data-driven model in capturing non-linearities, thus significantly enhancing the adaptability and robustness of the model during working condition migrations.

5.6.3. Interpretability analysis of the model decision-making process

The UKF fusion framework adopted in this study has advantages not only in improving accuracy but also in providing a dynamic and interpretable decision-making process. Different from black-box models, the UKF quantifies in real time the uncertainties of the physics-based model and the data model predictions through the Kalman gain matrix, and adaptively allocates the weights of the two in the final prediction accordingly. Its internal mechanism is as follows: during the stage when the confidence level of the physics-based model is high (such as in the early stage of wear), the system automatically assigns a higher weight to it to ensure the physical rationality of the prediction; when the nonlinearity of the process increases and the deviation of the physics-based model becomes larger (such as in the stage of severe wear), the weight of the data-driven model is dynamically increased for correction. This transparent fusion mechanism based on uncertainty clearly reveals the decision-making logic of the model and significantly enhances its credibility in industrial applications.

5.6.4. Analysis of computational efficiency

To evaluate the potential of the proposed method for application in industrial scenarios, this study analyzed the computational efficiency of the model on a standard hardware platform (Intel Xeon W-2265 CPU, NVIDIA GeForce RTX 3090 GPU, Python 3.11, PyTorch 2.1). The computational process of the model covers two stages: offline training and model execution.

During the offline training phase of the model, the first 240 groups of samples (dimension: 10000×6) out of all 300 groups of historical data are used to learn the parameters of the BiLSTM-AM model. The total time taken to complete 1000 epochs of training is approximately 80 min, and this process is regarded as a one-time computational investment. After the training is completed, the model parameters are fixed and can be directly used for subsequent prediction tasks. During the model execution phase, the total average time taken for a complete prediction of the remaining 60 sets of test samples (dimension: 10000×6) is approximately 560 ms. The specific calculation cost distribution is as follows: The data-driven model (forward inference of BiLSTM-AM) takes approximately 500 ms, the numerical solution of the dynamic model of the flank wear of the end mill takes approximately 25 ms, and the hybrid model takes approximately 35 ms to perform state fusion and update through the UKF algorithm.

Although this study did not achieve closed-loop online prediction on a real numerical control system, the above analysis shows that the complete prediction process has sub-second response capabilities on general-purpose computing hardware. This speed is much faster than the tool wear dynamic evolution process measured in minutes, which confirms the potential feasibility of this method to meet the real-time requirements of industrial online monitoring from the perspective of computing efficiency.

6. Conclusion

This research introduces a predictive approach for tool wear that combines a physics-based model with a data-driven model through the application of the UKF. This approach markedly improves the efficiency and accuracy of tool flank wear prediction, while also maintaining continuous machining operations. Its effectiveness and advantages are demonstrated through extensive comparative testing under diverse milling conditions. The key conclusions drawn from the study are as follows:

(1) The study leverages the state prediction and measurement update capabilities of UKF to efficiently integrate the data-driven model with the physics-based model, thereby enhancing the accuracy of tool wear prediction. The data-driven model, which is based on real-time monitoring signals, responds rapidly and sensitively to variations in the condition of tool wear, enabling instantaneous forecasting of wear. The physics-based model, formulated according to the mechanisms of abrasive, adhesive, and diffusive wear processes, offers stable long-term trend data. UKF dynamically integrates the strengths of both models, mitigating their respective limitations and effectively addressing the challenges posed by high nonlinearity and uncertainty in the tool wear process.
(2) Compared to traditional feature selection and dimensionality reduction methods, SCC is more effective in extracting sensitive features with a stronger nonlinear correlation to the tool wear value from the multi-domain feature set. This enhancement leads to a notable enhancement in the predictive precision of the data-driven model.
(3) Experimental data validation confirms that UKF can accurately estimate the actual state of tool wear. For different milling scenarios, it exhibits excellent predictive accuracy and robustness in contrast to individual prediction models. This makes it provide reliable decision support for tool maintenance and production process optimization.

Future research could further refine the physics-based model by incorporating additional factors such as milling vibration and oxidation wear, thereby improving the prediction accuracy to better meet practical requirements. Furthermore, the application of this integrated framework can be expanded to estimate RUL of tools, offering more comprehensive support for tool management and the optimization of production processes.

CRediT authorship contribution statement

Chunhua Feng: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Hui Ye:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Weidong Li:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Zhiwen Huang: Validation, Data curation. **Ming Li:** Writing – original draft, Validation, Formal analysis. **Jiaqi Chen:** Writing – original draft, Visualization, Formal analysis, Data curation. **Xun Xu:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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